

CHAPTER I

INTRODUCTION

1.1 OVERVIEW

The fact that solar electricity is abundant, together with lower costs of the conversion technology, has made it extremely popular. Solar energy is the conversion of light energy into electrical energy, often known as photovoltaic effect. Due to the use of solar electricity, pollution will be decreased and energy forecasts, homes, and communities may all be improved in addition to this. The monitoring of this system allows us to know about the system's condition, as well as telling us when anything is wrong, which is quite useful. Conventional sources like as coal, natural gas, and fossil fuels may be put to use again while renewable resources are re-utilized.. In this project, we have made a simple solar monitoring system by using an ESP8266 development board, where the solar panels are used for producing electricity, with the help of sunlight. The system is manually operated by human so, there is a need for an efficient approach that automatically control and monitor the current, voltage, and other parameters of the solar system and provides real-time statistics to users. The aim is to design solar energy monitoring and share information through IoT. It consists of a Arduino with an ESP8266 module, a voltage sensor, and a current sensor. When the ESP8266 module is a controller integrated with Wi-Fi, where the sensor checks the conditions that are programmed and detects the output voltage and current.

1.2 INTERNET OF THINGS (IOT)

The Internet of Things (IoT) describes the network of physical objects—“things”—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet. These devices range from ordinary household objects to sophisticated industrial tools. With more than 7 billion connected IoT devices today, experts are expecting this number to grow to 10 billion by 2020 and 22 billion by 2025. By means of low-cost computing, the cloud, big data, analytics, and mobile technologies, physical things can share and collect data with minimal human intervention. In this hyperconnected world, digital systems can record, monitor, and adjust each interaction between connected things. The physical world meets the digital world and they cooperate. While the idea of IoT has been in existence for a long time, a collection of recent advances in a number of different technologies has made it practical.

1.2.1 Access to low-cost, low-power sensor technology. Affordable and reliable sensors are making IoT technology possible for more manufacturers.

1.2.2 Connectivity. A host of network protocols for the internet has made it easy to connect sensors to the cloud and to other “things” for efficient data transfer.

1.2.3 Cloud computing platforms. The increase in the availability of cloud platforms enables both businesses and consumers to access the infrastructure they need to scale up without actually having to manage it all.

1.2.4 Machine learning and analytics. With advances in machine learning and analytics, along with access to varied and vast amounts of data stored in the cloud, businesses can gather insights faster and more easily. The emergence of these allied technologies continues to push the boundaries of IoT and the data produced by IoT also feeds these technologies.

1.2.5 Conversational artificial intelligence (AI). Advances in neural networks have brought natural-language processing (NLP) to IoT devices (such as digital personal assistants Alexa, Cortana, and Siri) and made them appealing, affordable, and viable for home use.

IoT Intelligent Applications are prebuilt software-as-a-service (SaaS) applications that can analyze and present captured IoT sensor data to business users via dashboards. We have a full set of IoT applications

IoT applications use machine learning algorithms to analyze massive amounts of connected sensor data in the cloud. Using real-time IoT dashboards and alerts, you gain visibility into key performance indicators, statistics for mean time between failures, and other information. Machine learning-based algorithms can identify equipment anomalies and send alerts to users and even trigger automated fixes or proactive counter measures.

With cloud-based IoT applications, business users can quickly enhance existing processes for supply chains, customer service, human resources, and financial services. There’s no need to recreate entire business processes.

1.3 INDUSTRIAL INTERNET OF THINGS (IIOT)

Industrial IoT (IIoT) refers to the application of IoT technology in industrial settings, especially with respect to instrumentation and control of sensors and devices that engage cloud technologies. Recently, industries have used machine-to-machine communication (M2M) to achieve wireless automation and control. But with the emergence of cloud and

allied technologies (such as analytics and machine learning), industries can achieve a new automation layer and with it create new revenue and business models. IIoT is sometimes called the fourth wave of the industrial revolution, or Industry 4.0.

The following are some common uses for IIoT:

- Smart manufacturing
- Connected assets and preventive and predictive maintenance
- Smart power grids
- Smart cities
- Connected logistics

The ability of IoT to provide sensor information as well as enable device-to-device communication is driving a broad set of applications. The following are some of the most popular applications and what they do.

Create new efficiencies in manufacturing through machine monitoring and product-quality monitoring.

Machines can be continuously monitored and analyzed to make sure they are performing within required tolerances. Products can also be monitored in real time to identify and address quality defects.

Improve the tracking and “ring-fencing” of physical assets.

Tracking enables businesses to quickly determine asset location. Ring-fencing allows them to make sure that high-value assets are protected from theft and removal.

Use wearables to monitor human health analytics and environmental conditions.

IoT wearables enable people to better understand their own health and allow physicians to remotely monitor patients. This technology also enables companies to track the health and safety of their employees, which is especially useful for workers employed in hazardous conditions.

Drive efficiencies and new possibilities in existing processes.

One example of this is the use of IoT to increase efficiency and safety in connected logistics for fleet management. Companies can use IoT fleet monitoring to direct trucks, in real time, to improve efficiency.

Enable business process changes.

An example of this is the use of IoT devices for connected access to monitor the health of remote machines and trigger service calls for preventive maintenance. The ability to remotely monitor machines is also enabling new product-as-a-service business models, where customers no longer need to buy a product but instead pay for its usage.

1.4 IOT CHANGING THE WORLD

The automotive industry stands to realize significant advantages from the use of IoT applications. In addition to the benefits of applying IoT to production lines, sensors can detect impending equipment failure in vehicles already on the road and can alert the driver with details and recommendations. Thanks to aggregated information gathered by IoT-based applications, automotive manufacturers and suppliers can learn more about how to keep cars running and car owners informed. IoT is reinventing the automobile by enabling connected cars. With IoT, car owners can operate their cars remotely—by, for example, preheating the car before the driver gets in it or by remotely summoning a car by phone. Given IoT's ability to enable device-to-device communication, cars will even be able to book their own service appointments when warranted.

The connected car allows car manufacturers or dealers to turn the car ownership model on its head. Previously, manufacturers have had an arms-length relationship with individual buyers (or none at all). Essentially, the manufacturer's relationship with the car ended once it was sent to the dealer. With connected cars, automobile makers or dealers can have a continuous relationship with their customers. Instead of selling cars, they can charge drivers usage fees, offering a "transportation-as-a-service" using autonomous cars. IoT allows manufacturers to upgrade their cars continuously with new software, a sea-change difference from the traditional model of car ownership in which vehicles immediately depreciate in performance and value.

1.4.1 Advantages of IoT

- I. It can assist in the smarter control of homes and cities via mobile phones. It enhances security and offers personal protection.
- II. By automating activities, it saves us a lot of time.
- III. Information is easily accessible, even if we are far away from our actual location, and it is updated frequently in real time.
- IV. Electric Devices are directly connected and communicate with a controller computer, such as a cell phone, resulting in efficient electricity use. As a result, there will be no unnecessary use of electricity equipment.
- V. Personal assistance can be provided by IoT apps, which can alert you to your regular plans.
- VI. It is useful for safety because it senses any potential danger and warns users. For example, GM OnStar, is a integrated device that system which identifies a car crash or accident on road. It immediately makes a call if an accident or crash is found.
- VII. It minimizes human effort because IoT devices connect and communicate with one another and perform a variety of tasks without the need for human intervention.
- VIII. Patient care can be performed more effectively in real time without the need for a doctor's visit. It gives them the ability to make choices as well as provide evidence-based care.
- IX. Asset tracking, traffic or transportation tracking, inventory control, delivery, surveillance, individual order tracking, and customer management can all be made more cost-effective with the right tracking system.

1.5 ROLE OF IOT IN SOLAR POWER SYSTEMS

The solar power monitoring device is important to stay informed about what the system is producing and how well it's performing. Picture a solar array on a roof or in a field that is made up of thousands of solar panels. All of them are wired together back to a mid-point, where solar inverters convert the direct current (DC) energy produced by the panels into alternating current (AC) energy which can be utilised to power building elements. To monitor several aspects of this process, most systems take the benefit of wired sensor networks.

The success of the monitoring system depends in part on the reliability of the communication network—how well the gateway can transfer information to the cloud. If for some reason the network fails, your devices won't be able to transmit data, which could cause you to miss important data indicating performance issues, or misinterpret other available data. Most all the monitoring systems have been comprised of wired sensors/components to date. While efficient, the cost and maintenance expense has restricted the number of sensors that one would place into service on a given system.

Efficiency proves to be a major hurdle in harnessing renewable energy. This especially holds true for methods that depend on variable resources like wind and solar energy. A technology-built IoT solution enables the implementation of automated controls to enhance efficiency with such methods of electricity production. With the help of these controls, energy from renewable resources can be harnessed with maximum efficiency. An IoT device can be useful to detect the most favorable conditions for energy production. The equipment can be adjusted accordingly to generate maximum output. The major benefit of IoT devices is that they generate data in real-time, which helps minimize wastage if any.

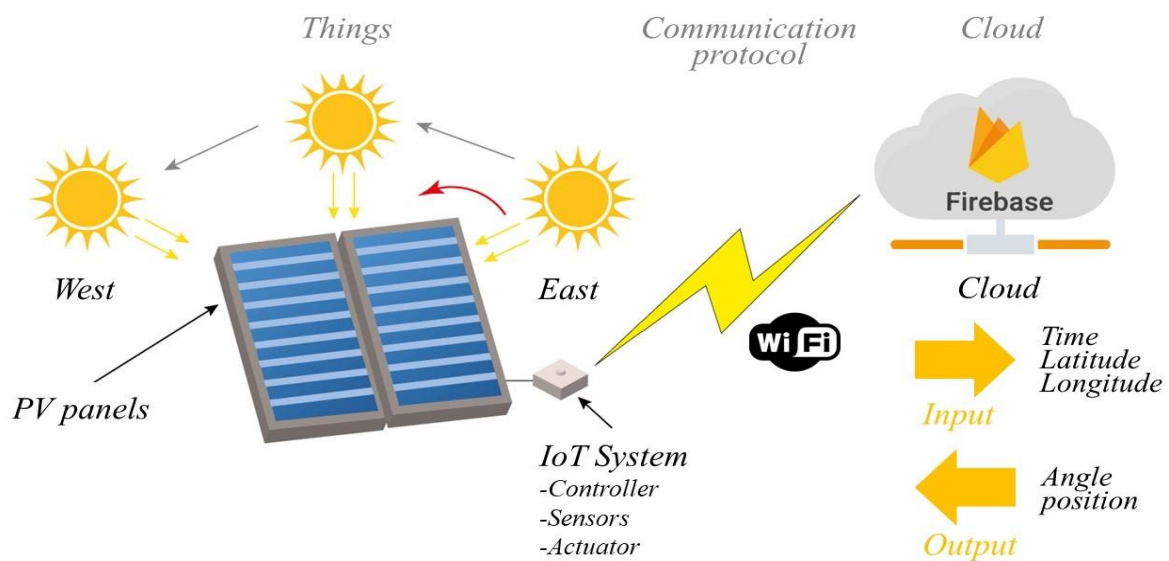


Fig 1.1 ROLE OF IOT IN PV SYSTEM

1.6 OBJECTIVE OF THE PROJECT

Solar power plants need to be monitored for optimum power output. This helps retrieve efficient power output from power plants while monitoring for faulty solar panels, connections, dust accumulated on panels lowering output and other such issues affecting solar performance. The main objective of this proposed work is to monitor the output of PV system using the current and voltage value sensed by the arduino. To implement in smart grid, this system helps for efficient usage. IoT based wireless systems are widely chosen in order to avoid associated risks with wired systems. While keeping in mind the needs of near future, where every device will be smart, automated and connected via internet.

1.7 STRUCTURE OF THE PROJECT

The main intention of this proposed project is to get maximum power output from the solar panels. Additionally, if there is any improper functioning of the solar panels will be shown and also the parameters like voltage and current are monitored by using the sensors and displayed by using the IoT technology. This model is explained by using the solar radiation i.e., sunlight from the sun is trapped by the solar panels and then these solar panels capture sunlight and turn into useful energy forms of energy such as heat and electricity. Then the obtained electrical energy is sensed by the sensors such as voltage sensor sense the voltage generated by the solar panel with the help of voltage divider principle and current is obtained by using mathematical formulation. The experimental arrangement of the introduced system consists of solar panels, Regulator power supply, Wi-Fi module ESP8266.

1.8 WORK PLAN OF THE PROJECT

In this project the work plan begins with the literature review and mainly focus on the IOT based projects which are related to the PV monitoring system. After the completion of literature survey the work plan moves on to the next stage which includes testing of sensors, modules and other components which are required for the project. After the successful completion of testing the hardware components the work plan shifts to the next level, designing and development of power circuit and interfacing the output to the WIFI module. After the implementation of power circuit the work plan comes to final stage of testing the final output with the IOT platform and monitoring the output parameters.

1.9 SUMMARY

The above chapter gives the overall information about the Internet of things and Industrial Internet of Things and the benefits of using the IOT related objects and advantages. The concept behind the project mainly focus on the IOT based PV monitoring system which require several hardware and a software to be implemented. Hardware components includes sensors, Analog to digital converters, Microprocessor, WiFi modules and a cloud. After the implementation of this project the PV paramteres can be monitored wireless via cloud website and even through Android app. IoT-based remote monitoring will increase the system's energy efficiency by utilizing low-power sophisticated wireless modules, consequently lowering the carbon footprint. The use of a Web Console-based interface will greatly save manual supervision time and aid in the scheduling of plant management tasks.

CHAPTER II

LITERATURE REVIEW

2.1 REVIEWS OF LITERATURE SURVEY

Kekre.A & Gawre S.K [1] have presented the low cost Pv monitoring system .As technology is advancing, cost of renewable energy equipments is decreasing which has resulted in a massive increase in solar photovoltaic installations. Most of these installations act as auxiliary power source. A majority of these are installed in inaccessible locations — as close as a rooftop to as far away as a dessert. Hence they require sophisticated systems for remote monitoring of these installations using wide area networks. In this paper we will discuss a low cost IOT based embedded Solar PV Monitoring system which will make use of GPRS module and a low cost microcontroller to send the data measured at the production end on the internet, which can then be accessed anywhere on the globe. This will provide us real time information of the installation which will help us in its maintenance, fault detection and will give us a record of all the data at fixed intervals.

Adhya, S., Saha, D., Das, A., Jana, J.,& Saha, H [2] have implemented the evaluation of performance monitoring in using the Internet Of Things Technology. For supervising solar photovoltaic power generation can greatly enhance the performance, monitoring and maintenance of the plant. With advancement of technologies the cost of renewable energy equipments is going down globally encouraging large scale solar photovoltaic installations. This massive scale of solar photovoltaic deployment requires sophisticated systems for automation of the plant monitoring remotely using web based interfaces as majority of them are installed in inaccessible locations and thus unable to be monitored from a dedicated location. The discussion in this paper is based on implementation of new cost effective methodology based on IoT to remotely monitor a solar photovoltaic plant for performance evaluation. This will facilitate preventive maintenance, fault detection, historical analysis of the plant in addition to real time monitoring.

Shrihariprasath, B., & Rathinasabapathy, V. [3] have created an effective implementation of an intelligent remote monitoring system for solar Photovoltaic (PV) Power Conditioning Unit (PCU) which is used in a greenhouse environment. The proposed system design can be installed in solar PV PCU in order to solve management problems, maintenance and shortens

the mean time to repair. We have designed a smart remote monitoring system based on internet of thing for monitoring Solar PV PCU. This system had incorporated remote monitoring for solar PV PCU through internet using host, network GPRS (Global Positioning Radio Service), embedded system gateway and other components. The result of our demonstration shows that the system can monitor store and manipulate data from solar PV PCU. Thus, the remote monitoring functions are realized in real-time.

Patil, S., Vijayalashmi, M., & Tapaskar, R. [4] have built a framework using Raspberry pi for online display. The Internet of Things has a vision in which the internet extends into the real world, which incorporates everyday objects. The IoT allows objects to be sensed or controlled remotely over existing network infrastructure, creating opportunities for pure integration of the physical world into computer-based systems, and resulting in improved efficiency, accuracy and economic benefit in addition to reduced human intervention. This technology has many applications like Solar cities, Smart villages, Micro grids and Solar Street lights and so on. As Renewable energy grew at a rate faster than any other time in history during this period. The proposed system refers to the online display of the power usage of solar energy as a renewable energy. This monitoring is done through raspberry pi using flask framework. Smart Monitoring displays daily usage of renewable energy. This helps the user to analysis of energy usage. Analysis impacts on the renewable energy usage and electricity issues.

Phung, M. D., De La Villefromoy, M., & Ha, Q [5] have demonstrated the control system based IOT for Solar energy generation requires efficient monitoring and management in moving towards technologies for net-zero energy buildings. This paper presents a dependable control system based on the Internet of Things (IoT) to control and manage the energy flow of renewable energy collected by solar panels within a microgrid. Data for optimal control include not only measurements from local sensors but also meteorological information retrieved in real-time from online sources. For system fault tolerance across the whole distributed control system featuring multiple controllers, dependable controllers are developed to control and optimise the tracking performance of photovoltaic arrays to maximally capture solar radiation and maintain system resilience and reliability in real time despite failures of one or more redundant controllers due to a problem with communication, hardware or cybersecurity. Experimental results have been obtained to evaluate the validity of the proposed approach.

Katyarmal, M., Walkunde, S., Sakhare, A., & Rawandale, U. S.[6] have presented the Journal of IOT based remote monitoring. The power generated from solar panel is to be efficiently monitored and managed to reduce the generation losses in solar power generation. Generally, we use solar plants to build in the locations where people can't reach on daily basis so this approach will help them to virtually control their systems from faraway places. It monitors the panel load by using the IoT technologies and the data which are received from the panels are send to the cloud through the internet for the future use. It also helps the remote users to monitor the solar power plant. The user can get the information about the current and previous average parameter like voltage, temperature and current. This will facilitate the fault detection and preventive maintenance of solar. In this paper we use the application Internet of thing (IOT) to control and monitor the solar power (renewable energy). This system is designed to solve the problem occur in solar power generation like management problem, maintenance and to reduce the time of repair. Using this technology, the cost of solar energy (renewable energy) generation reduces. This also provide real time information to the user help to monitor the system. The main purpose of this paper is that the solar panel can collect or capture maximum solar radiation and maintain the system more reliably and efficiently.

Purusothaman, S. D., Rajesh, R.,Bajaj, K. K., & Vijayaraghavan, V. [7] have demonstrated the realtime monitoring using VLSI system. An embedded system is characterized by real-time and multi-rate operations define the ways in which the system works, reacts to events and interrupts and schedules the system's functioning in real time. Typically, the end user does not develop new software for the embedded device. With advances in VLSI technology, embedded systems have become so inexpensive that they are found in most of today's electronic devices. An implementation of a multi-agent system (MAS) for a rural Indian micro grid (MG) using Arduino has been explained in this paper. The focus is on the DG agents- the distributed energy resources (DERs), load, storage and the grid agents - and the Mu agent which acts as the communication channel between the DG agents to the higher level agents such as he control agent. The design philosophy incorporates the core features of an MAS that are required to monitor and control a rural Indian microgrid. The system has been compacted and design simplified, since only relevant parameters from a rural Indian MG are considered.

M. N. Akram and S. Lotfifard [8] have performed a standard operating test condition on PV systems. A health monitoring method for photovoltaic (PV) systems based on probabilistic neural network (PNN) is proposed that detects and classifies short- and open-circuit faults in real time. To implement and validate the proposed method in computer programs, a new approach for modeling PV systems is proposed that only requires information from manufacturers datasheet reported under normal-operating cell temperature (NOCT) conditions and standard-operating test conditions (STCs). The proposed model precisely represents characteristics of PV systems at different temperatures, as the temperature dependency of parameters such as ideality factor, series resistance, and thermal voltage is considered in the proposed model. Although this model can be applied to a variety of applications, it is specifically used to test and validate the performance of the proposed fault detection and classification method.

Alrwashdeh, SAAD S. [9] have designed the tracking system in PV system according to solar irradiance. Saudi Arabia is one of the countries of the world rich in natural resources, foremost of which is solar energy with average solar radiation of between 4 and 7 KWh/m². This scientific work examines the best ways to exploit solar energy in Saudi Arabia to reduce dependence on traditional energy sources, especially as the prices of energy produced from fossil fuels and renewable energy systems, particularly solar systems, have become close. For that, a PV design with a fixed tilt angle and two tracking systems is presented in this work in order to select the best way to lead to maximizing energy production.

Al-Fuqaha, M. Guizani, M. Mohammadi, M. Aledhari, M. Ayyash [10] have built an smart IOT based machine to machine application. An overview of the Internet of Things (IoT) with emphasis on enabling technologies, protocols, and application issues. The IoT is enabled by the latest developments in RFID, smart sensors, communication technologies, and Internet protocols. The basic premise is to have smart sensors collaborate directly without human involvement to deliver a new class of applications. The current revolution in Internet, mobile, and machine-to-machine (M2M) technologies can be seen as the first phase of the IoT.

2.2 SUMMARY

After the completion of literature survey we have concluded that IOT based PV monitoring system has efficient low cost cloud storage system. The sensors which are

implemented in the project are low power consumption, high precision and accuracy. The main purpose of this literature review is that the solar panel can collect or capture maximum solar radiation and maintain the system more reliably and efficiently

CHAPTER III

EXISTING REGIME

3.1 SOLAR PANEL OPERATION AND MAINTENANCE

In today's current scenario, industrial and commercial establishments are resorting to solar arrays and industrial PV systems in order to bypass power outage problems associated with grid supply. Be it in the paper, steel, chemical, textile, cement, dairy or the ceramic industry, solar power panels are fast becoming a norm rather than an option in India. This has been given a boost by the presence of open access grids in most states, nationwide.

Most solar panel operation and maintenance (O&M) processes require periodic checks for ensuring optimal performance and security. The frequency and degree of solar array monitoring and solar panel maintenance are reliant on the installation type, system configuration and location of the commercial solar power system in use.

Yearly inspections are recommended by companies providing solar maintenance in India. Apart from inspecting PV systems regularly, it is also important to carry out inspections when their ownership changes.

Solar panels typically don't require much maintenance other than periodic cleaning and keeping them free from obstacles that can cast shadows over the panels. Solar panels need an unobstructed path to the sun to operate optimally.

3.2 INVOLVEMENT OF TECHNICIAN

Solar PV systems' maintenance processes are best carried out by qualified installers or service technicians who are conversant with PV components and their safety procedures. These providers of solar panel maintenance services take care of the full range of PV maintenance processes that include:

- I. Carrying out visual inspections
- II. Conducting verifications of the PV system operations
- III. Taking corrective actions

- IV. Monitoring and verifying how effective the corrective actions are, etc.

3.2.1 PROCESS INVOLVED IN THE MAINTENANCE OF SOLAR POWER SYSTEM

Popular packages encompass processes involved in the maintenance of solar panels and solar power plants; such as:

- I. The timely and regular cleaning of solar cells and PV panels
- II. Regular maintenance of all thermal-based components
- III. Servicing of HT side equipment on an annual basis
- IV. Diagnosis and tests pertaining to low solar power production
- V. Testing and upkeep of circuits
- VI. Tracing of IV curves and thermal imaging
- VII. Measure of earth value
- VIII. Retro commissioning
- IX. Management of warranties
- X. Systems check pertaining to data acquisition, etc

3.2.2 SERVICES PROVIDED BY SOLAR PANEL MAINTENANCE COMPANIES

As far as solar PV systems' uptime management is concerned, the companies involved in solar panel operation and maintenance offer:

- I. Data interpretation and collection of data in real time
- II. Monitoring of energy production on a consistent basis
- III. Critical as well as non-critical repairs
- IV. Analysis of trends and KPIs in order to ensure continuous improvement
- V. Selection of specific key performance indicators to minimize cost while delivering the best possible system performance in the face of varying conditions.
- VI. Management of inventory, spare parts, etc.
- VII. The tracking of solar power generation logs and updating document service histories

- VIII. Optimization of balance between yield, cash flows and costs of scheduled maintenance, etc.
- IX. Remote or on-site sensing of the various environmental conditions impacting the PV system being maintained.
- X. Assessment of components and inverters to ascertain whether they have to be repaired or restored.
- XI. Identification of safety issues, etc.

Companies taking up the responsibility of monitoring the operations and managing solar rooftops' upkeep provide 24/7 customer support via their network operations centre. They offer in-depth business insights in the form of value-added services to their corporate customers and industry-based client.

3.2.3 COST INVOLVED IN SOLAR PANEL MAINTENANCE

Commercial solar system maintenance costs are minuscule in comparison to the many benefits of lighting, ventilation, equipment power supply, etc. derived by installing them. That said, the costs of solar PV maintenance and operations are dependent on:

- I. Location: controlled access, remote or for restricted hours of operations
- II. Type of solar panel system: ground mounted, roof-top, fixed vs. tracking, etc.
- III. PV components: number of combiners and solar modules, type and number of inverters, number of transformers, etc.
- IV. Cover provided by the warranties in place: environmental conditions like salt air, humidity, heat, pollen, sandy location, population of birds, industrial emissions, heavy insolation, presence of construction sites in proximate locations, etc.

Given the scenario of heavy usage and high electricity bills, in-house solar power plants in the form of captive models and/or off-site solar farms are becoming the first choice for industries. These solar energy producing units are useful during times of peak loading of grids, which result in power outages. Additionally, they eliminate the high tariff rates applicable at peak hours.

3.3 PHYSICAL STORAGE FOR DATA ACQUISITION

A microcontroller based data acquisition system used to collect data from panel. The system comprises of data acquisition system, Zigbee and RS-232 communication modules

and computer that permit user to collect data wirelessly. 8-bit 40-pin Atmel AT89C51 type microcontroller used to monitor. After collecting data wireless communication module Zigbee and serial communication module RS232 used to transmit collected data to user computer.

Characteristics and operation of setup complete system provided and result of this experiment show that data acquisition is simple and stable.

3.3.1 Microcontroller AT89C51: AT89C51, Atmel designed 8-bit microcontroller that has 4K bytes of flash programmable and erasable read only memory (PEROM). This microcontroller is compatible with MCS-51 instruction set and pinout. Operation frequency varies from 0 to 24MHz. 11.0592MHz crystal frequency set to currently so timing delay is 1.085 μ sec.

3.3.2 Analog to Digital Converter ADC0831: ADC0831 is 8-bit serial data output 8 pin analog to digital converter IC that converts continuous analog signal to a digital number form. ADC0831 converter used with microcontroller to convert analog data received from solar panel to digital form to display or store.

3.3.3 Communication Modules Zigbee and RS232: Zigbee is an IEEE 802.15 standard based communication protocol used to create wireless personal area network between solar panel and host computer to transmit data to host system. RS232 is a serial communication standard used for transmission of data between data terminal equipment as host computer and data circuit terminating or communication equipment Zigbee. MAX232 IC used to complete these actions and it use only a 5V voltage can be TTL level converter for RS232.

3.3.4 Keil: The μ vision integrated development environment (IDE) from keil provides support for embedded software development. To instruct microcontroller AT89C51, this one used to decide controls. It provides facility to editing source code, debug programs and also complete simulation in powerful environment. 1456 Sunil Parashar and Sunil Dhankhar

3.3.5 Proteus: Proteus virtual system modeling is simulation tool that is a combination of SPICE circuit simulation, different animated components and various microprocessor based design, used to simulate and test design before create physical prototype. Complete acquisition model designed into this environment to check its complexity level. Availability of serial communication allows it to transmission of data to other modules.

3.3.6 Visual Basic 6.0: Visual basic is an event driven language with an integrated development environment allow it to perform some special functionality as rapid application development (RAD) of graphical user interface, database access and remote data objects etc. It helps to gather data in database and create a graph or chart to make comparison easy and understandable.

3.3.7 Data Acquisition System Model: The data acquisition system software developed using embedded C language. Compiler tool to compile this code is keil C51 compiler and block diagram of system software shown in figure 2. Initialization of complete system by sending control or data bits, process data acquisition means access data and forward to host system or storage media through any communication modules as Zigbee and RS-232.

Data acquisition system is a process used to collect information that can be stored and processed by a computer to analyze some special phenomenon. Two main aspects are essential for this system: transmission type in system (serial or parallel) and information exchange mode between system devices. The acquisition system is essentially based on Atmel AT89C51 and ADC0831 components.

When system start first of all continuous analog data send to analog to digital converter that accept data over and generates data in digital format. This generated data is now sending to any pin of controller to allow controller to perform action according incoming value. After processing, data is transmitted to host system through wireless communication module Zigbee and serial communication RS-232. Now data is collected at host system in a database and graph created in Visual basic module to understand changes.

This data acquisition system makes use of microcontroller hardware and software modules so it can be easier to reduce complexity of entire system in terms of hardware and software designs. Maintenance cost and upgradation cost of system is high because of use of microcontroller.

3.4 SUMMARY

The above existing regime states that the PV monitoring system comprises of high maintenance cost and labour cost. Solar panel typically don't require any maintenance rather than periodic cleaning and keeping free from obstacle. The data acquisition system uses physical storage devices such as microprocessor and hard disc. Two main aspects are

essential for this system: transmission type in system (serial or parallel) and information exchange mode between system devices. The acquisition system is essentially based on Atmel AT89C51 and ADC0831 components.

CHAPTER IV

IMPLEMENTATION OF IOT BASED PHOTOVOLTAIC PARAMETER MONITORING SYSTEM USING NODEMCU ESP8266

4.1 OVERVIEW

The real-time solar energy monitoring system is proposed based on the three-layer architecture of Internet of Things (IoT). The three-stage architecture is shown in Fig.. The lower layer contains sensing and actuating devices like sensors, actuators, and controllers since it is a combination of sensing and processing layer. The next layer is a middle layer which encompasses network layer with wired and wireless network like LAN, 4G, Wi-Fi etc., act as a gateway to route the packets (data) to the transport layer that contains TCP/IP, UDP, for further transmission of data to the upper end. The final stage is the application layer deliver user interface and cloud platform for remote access.

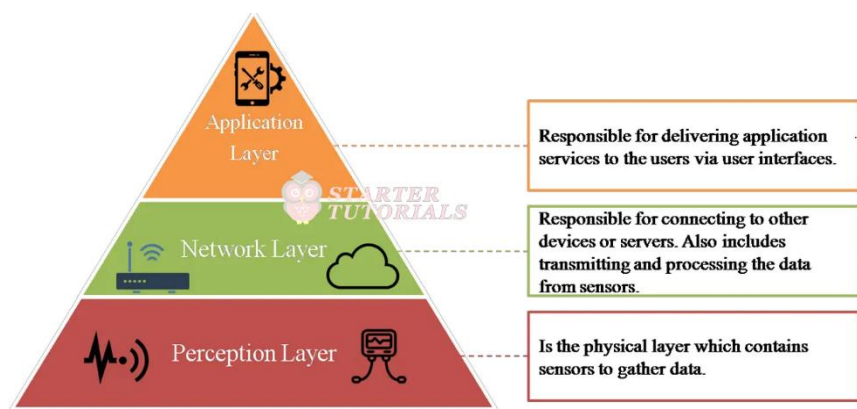


Fig.4.1 Three Layer Architecture of Internet of Things

The block diagram of IoT based solar monitoring system is shown in Fig.. This illustrates the outline of our proposed work. Poly Crystalline silicon of 5-watt solar panel is used for a monitoring system. The voltage and current sensors are used to measure the respective voltage and current from the panel. The temperature sensor is placed on the solar PV module

to measure the current temperature which greatly affects the efficiency of the solar panel. Light Dependent Resistor is used to measure the amount of solar irradiance in a planar surface in terms of lux . The Arduino Mega 2560 and NodeMCU ESP8266 plays a pivotal role in handling the measured data for processing and forwards the data to the cloud platform through Wi-Fi module for concurrent observation and decision making.

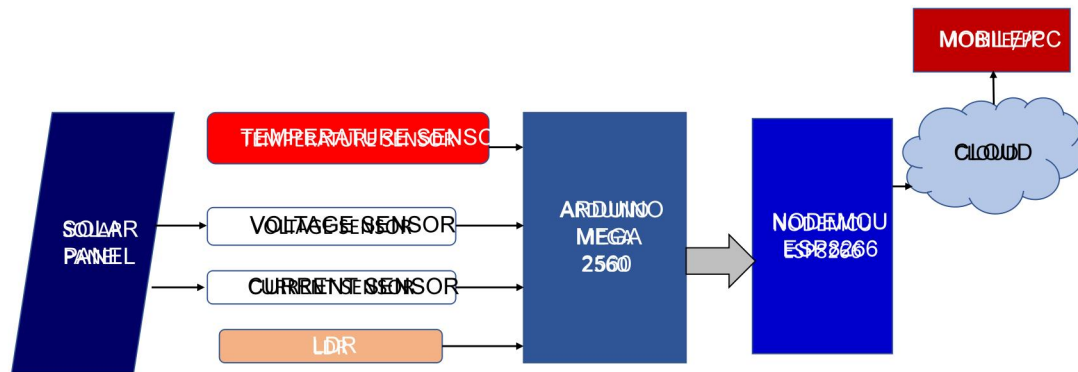


Fig.4.2. PROPOSED BLOCK DIAGRAM OF IOT BASED PV MONITORING SYSTEM

4.2 REQUIREMENTS FOR PROJECT

4.2.1 SOLAR PANEL

A solar panel is a device that converts sunlight into electricity by using photovoltaic (PV) cells. PV cells are made of materials that generate electrons when exposed to light. The electrons flow through a circuit and produce direct current (DC) electricity, which can be used to power various devices or stored in batteries. Solar panels are also known as solar cell panels, solar electric panels, or PV modules.

4.2.2 RATINGS OF SOLAR PANEL:

RATED MAXIMUM POWER (P_{max}) = 5W

RATED OPERATING VOLTAGE (V_{max}) = 8V

RATED OPERATING CURRENT (I_{max}) = 0.59A

OPEN CIRCUIT VOLTAGE (V_{oc}) $=10V$

SHORT CIRCUIT CURRENT (I_{oc}) $=0.67A$



Fig.4.3 SOLAR PANEL – 5W

4.2.3 VOLTAGE SENSOR

Generally, a sensor is an electrical device used to detect as well as respond to a particular type of signal like optical or electrical. Implementation of sensor techniques in voltage or current has become an outstanding option toward the measurement of voltage & current methods. The advantages of sensors over conventional methods for measuring mainly include less size and weight, high safety, high accuracy, nonsaturable, eco-friendly, etc. It is feasible to merge both the current and voltage measurement into a physical device with tiny and solid dimensions. This article discusses an overview of the voltage sensor and its working.

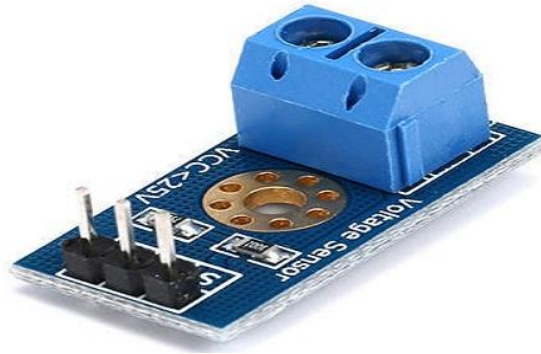


Fig 4.4 Voltage Sensor

This sensor is used to monitor, calculate and determine the voltage supply. This sensor can determine the AC or DC voltage level. The input of this sensor can be the voltage whereas the output is the switches, analog voltage signal, a current signal, an audible signal, etc. Some sensors provide sine waveforms or pulse waveforms like output & others can generate outputs like AM (Amplitude Modulation), PWM (Pulse Width Modulation) or FM (Frequency Modulation). The measurement of these sensors can depend on the voltage divider.

This sensor includes input and output. The input side mainly includes two pins namely positive and negative pins. The two pins of the device can be connected to the positive & negative pins of the sensor. The device positive & negative pins can be connected to the positive & negative pins of the sensor. The output of this sensor mainly includes supply voltage (Vcc), ground (GND), analog o/p data.

4.3.Types of Voltage Sensors

These sensors are classified into two types like a resistive type sensor and capacitive type sensor.

4.3.1 Resistive Type Sensor

This sensor mainly includes two circuits like a voltage divider & bridge circuit. The resistor in the circuit works as a sensing element. The voltage can be separated into two resistors like a reference voltage & variable resistor to make a circuit of the voltage divider. A voltage supply is applied to this circuit. The output voltage can be decided by the resistance used in the circuit. So the voltage change can be amplified.

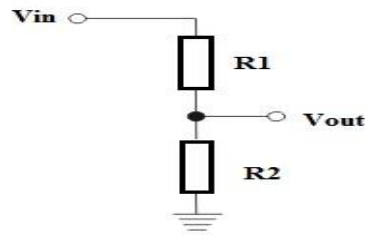


Fig 4.5 Resistive Type Sensor

The bridge circuit can be designed with four resistors. One of these resistors can be subjected to the voltage detector device. The change in voltage can be directly exhibited. This difference alone can be amplified but the difference within the voltage divider circuit not only amplified.

$$V_{out} = (R1/R1 + R2) * V_{in}$$

4.3.2 Capacitor Type Sensor

This type of sensor consists of an insulator and two conductors within the center. As the capacitor is power-driven with 5 Volt, then the flow of current will be there in the capacitor. This can create revulsion of electrons within the capacitor. The difference in capacitance indicates the voltage and the capacitor can be connected within the series.

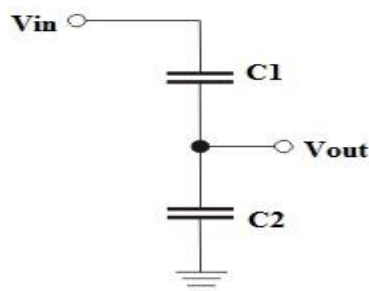


Fig 4.6 Capacitive Type sensor

4.3.3 Applications

The applications of this sensor include the following.

- I. Detection of power failure
- II. Detecting of load
- III. Safety switching

- IV. Controlling temperature
- V. Controlling of power demand
- VI. Detection of fault
- VII. Variation of load measurement of Temperature

Thus, this is all about voltage sensor which can be used to detect the range of voltage in any device. It decides the electrical charge within any device. The working principle of this sensor mainly depends on the principle of either capacitive or resistive.

4.4 ACS712 Current Sensor

Current flowing through a conductor causes a voltage drop. The relation between current and voltage is given by Ohm's law. In electronic devices, an increase in the amount of current above its requirement leads to overload and can damage the device.

Measurement of current is necessary for the proper working of devices. Measurement of voltage is Passive task and it can be done without affecting the system. Whereas measurement of current is an Intrusive task which cannot be detected directly as voltage. For measuring current in a circuit, a sensor is required. ACS712 Current Sensor is the sensor that can be used to measure and calculate the amount of current applied to the conductor without affecting the performance of the system.

ACS712 Current Sensor is a fully integrated, Hall-effect based linear sensor IC. This IC has a 2.1kV RMS voltage isolation along with a low resistance current conductor.

4.4.1 Working Principle

Current Sensor detects the current in a wire or conductor and generates a signal proportional to the detected current either in the form of analog voltage or digital output.

Current Sensing is done in two ways – Direct sensing and Indirect Sensing. In Direct sensing, to detect current, Ohm's law is used to measure the voltage drop occurred in a wire when current flows through it.

A current-carrying conductor also gives rise to a magnetic field in its surrounding. In Indirect Sensing, the current is measured by calculating this magnetic field by applying either Faraday's law or Ampere law. Here either a Transformer or Hall effect sensor or fiberoptic current sensor are used to sense the magnetic field.

ACS712 Current Sensor uses Indirect Sensing method to calculate the current. To sense current a liner, low-offset Hall sensor circuit is used in this IC. This sensor is located at the surface of the IC on a copper conduction path. When current flows through this copper conduction path it generates a magnetic field which is sensed by the Hall effect sensor. A voltage proportional to the sensed magnetic field is generated by the Hall sensor, which is used to measure current. The proximity of the magnetic signal to the Hall sensor decides the accuracy of the device. Nearer the magnetic signal higher the accuracy. ACS712 Current Sensor is available as a small, surface mount SOIC8 package. In this IC current flows from Pin-1 and Pin-2 to Pin-3 and Pin-4. This forms the conduction path where the current is sensed. Implementation of this IC is very easy.

ACS712 can be used in applications requiring electrical isolation as the terminals of the conduction path are electrically isolated from the IC leads. Thus, this IC doesn't require any other isolation techniques. This IC requires a supply voltage of 5V. Its output voltage is proportional to AC or DC current. ACS712 has a nearly zero magnetic hysteresis.

Where Pin-1 to Pin-4 forms the conduction path, Pin-5 is the signal ground pin. Pin-6 is the FILTER pin that is used by an external capacitor to set the bandwidth. Pin-7 is the analog output pin. Pin-8 is the power supply pin.

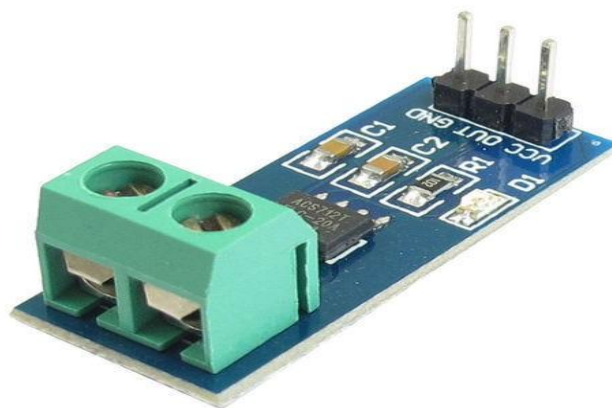


Fig 4.7 ACS712 CURRENT SENSOR

4.4.2 Applications of ACS712 Current Sensor

This IC can detect both AC and DC current so, it has a wide range of applications. ACS712 is used in Peak detection circuits, circuits to increase gain, rectification application for AtoD converters, Overcurrent fault latch, etc...The filter pin provided by this IC is used to eliminate the attenuation effect in resistor divider circuits.

ACS712 is used in many industrial, commercial and communication applications. This IC is applicable for Automobile applications. Some of the typical applications of this IC can be found in motor control circuits, for load detection and management, SMPS, overcurrent fault protection circuit.

This IC can measure current for high voltage loads operating at 230V AC mains. To read the values it can be easily interfaced with the ADC of a microcontroller.

4.5 DHT11 Sensor

Humidity is the measure of water vapour present in the air. The level of humidity in air affects various physical, chemical and biological processes. In industrial applications, humidity can affect the business cost of the products, health and safety of the employees. So, in semiconductor industries and control system industries measurement of humidity is very important. Humidity measurement determines the amount of moisture present in the gas that can be a mixture of water vapour, nitrogen, argon or pure gas etc... Humidity sensors are of two types based on their measurement units. They are a relative humidity sensor and Absolute humidity sensor. DHT11 is a digital temperature and humidity sensor.

DHT11 is a low-cost digital sensor for sensing temperature and humidity. This sensor can be easily interfaced with any micro-controller such as Arduino, Raspberry Pi etc... to measure humidity and temperature instantaneously.

DHT11 humidity and temperature sensor is available as a sensor and as a module. The difference between this sensor and module is the pull-up resistor and a power-on LED. DHT11 is a relative humidity sensor. To measure the surrounding air this sensor uses a thermistor and a capacitive humidity sensor.

4.5.1 Working Principle of DHT11 Sensor

DHT11 sensor consists of a capacitive humidity sensing element and a thermistor for sensing temperature. The humidity sensing capacitor has two electrodes with a moisture holding substrate as a dielectric between them. Change in the capacitance value occurs with the change in humidity levels. The IC measure, process this changed resistance values and change them into digital form.

For measuring temperature this sensor uses a Negative Temperature coefficient thermistor, which causes a decrease in its resistance value with increase in temperature. To get larger resistance value even for the smallest change in temperature, this sensor is usually made up of semiconductor ceramics or polymers.

The temperature range of DHT11 is from 0 to 50 degree Celsius with a 2-degree accuracy. Humidity range of this sensor is from 20 to 80% with 5% accuracy. The sampling rate of this sensor is 1Hz .i.e. it gives one reading for every second. DHT11 is small in size with operating voltage from 3 to 5 volts. The maximum current used while measuring is 2.5mA. DHT11 sensor has four pins- VCC, GND, Data Pin and a not connected pin. A pull-up resistor of 5k to 10k ohms is provided for communication between sensor and micro-controller.

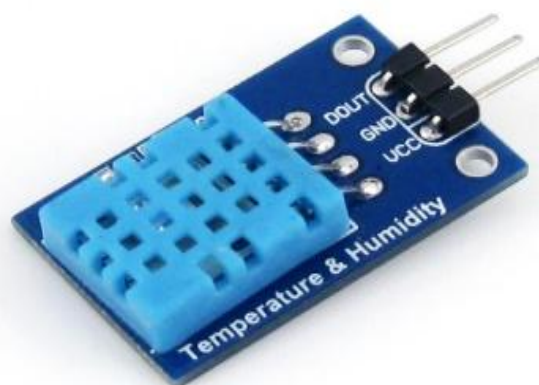


Fig 4.8 DHT 11 Sensor

4.5.2 Applications

This sensor is used in various applications such as measuring humidity and temperature values in heating, ventilation and air conditioning systems. Weather stations also use these sensors to predict weather conditions. The humidity sensor is used as a preventive measure in homes where people are affected by humidity. Offices, cars, museums, greenhouses and industries use this sensor for measuring humidity values and as a safety measure.

It's compact size and sampling rate made this sensor popular among hobbyists. Some of the sensors which can be used as an alternative to DHT11 sensor are DHT22, AM2302, SHT71.

4.6 LIGHT DEPENDENT RESISTOR

An electronic component like LDR or light-dependent resistor is responsive to light. Once light rays drop on it, then immediately the resistance will be changed. The resistance values of an LDR may change over several orders of magnitude. The resistance value will be dropped when the light level increases.

The resistance values of LDR in darkness are several megaohms whereas in bright light it will be dropped to hundred ohms. So due to this change in resistance, these resistors are extremely used in different applications. The LDR sensitivity also changes through the incident light's wavelength.

The designing of LDRs can be done by using semiconductor materials to allow their light-sensitive properties. The famous material used in this resistor is CdS (cadmium sulfide), even though the utilization of this material is currently restricted in European countries due to some environmental issues while using this material. Likewise, CdSe (cadmium selenide) is also restricted and additional materials that can be employed mainly include PbS (lead sulfide), InS (indium antimonide).

Even though for these resistors, a semiconductor material is used, because they are simply passive devices and they do not have a PN-junction. This detaches them from other LDRs such as phototransistors & photodiodes.

4.6.1 Working Principle of Light Dependent Resistor

The working principle of an LDR is photoconductivity, which is nothing but an optical phenomenon. When the light is absorbed by the material then the conductivity of the material enhances. When the light falls on the LDR, then the electrons in the valence band of the material are eager to the conduction band. But, the photons in the incident light must have energy superior to the bandgap of the material to make the electrons jump from one band to another band (valance to conduction). Hence, when light having ample energy, more electrons are excited to the conduction band which grades in a large number of charge carriers. When the effect of this process and the flow of the current starts flowing more, the resistance of the device decreases.

4.6.2 Light Dependent Resistor Specifications

The LDR specifications mainly include maximum power dissipation, maximum operating voltage, peak wavelength, dark resistance, etc. The values of these specifications mentioned below.

- I. Maximum power dissipation is 200mW
- II. The maximum voltage at 0 lux is 200V
- III. The peak wavelength is 600nm
- IV. Minimum resistance at 10lux is $1.8k\Omega$
- V. Maximum .resistance at 10lux is $4.5k\Omega$
- VI. Typical resistance at100lux is $0.7k\Omega$
- VII. Dark resistance after 1 sec is $0.03M\Omega$
- VIII. Dark resistance after 5 sec is $0.25M\Omega$

4.6.3 Types of LDR

The classification of LDR can be done based on linearity or photosensitive materials. The most frequently used materials while designing LDR mainly include cadmium sulfide, thallium sulfide, cadmium selenide & lead sulfide.

- I. The LDRs generated with this cadmium sulfide chemical are very responsive to all kinds of light radiation that are noticeable within the spectrum of human beings.
- II. The LDRs made with lead sulfate chemicals are particularly responsive to IR radiation.

The most frequent classification of LDR is linear as well as nonlinear

- I. Linear type LDRs are called as photodiodes however in some applications areas it is used as photoresistors because of the linear performance they present & their operation.
- II. Non-linear LDRs are commonly used but their behavior does not depend on the polarity through which it unites.

4.6.4 LDR Advantages

The advantages of LDR include the following.

- I. Sensitivity is High
- II. Simple & Small devices
- III. Easily used
- IV. Inexpensive
- V. There is no union potential.
- VI. The light-dark resistance ratio is high.
- VII. Its connection is simple

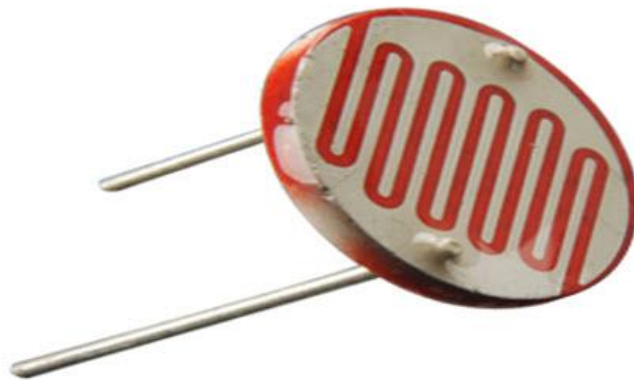


Fig 4.9 Light Dependent Resistor

4.6.5 Applications of LDR

Light-dependent resistors are simple and low-cost devices. These devices are used where there is a need to sense the presence and absence of light is necessary. These resistors are used as light sensors and the applications of LDR mainly include alarm clocks, street lights, light intensity meters, burglar alarm circuits. For a better understanding of this concept, here we have explained one project namely; power conserving of intensity controlled street lights using LDR.

4.7 ARDUINO MEGA 2560

Arduino board is an open-source microcontroller board which is based on Atmega 2560 microcontroller. The growth environment of this board executes the processing or wiring language. These boards have recharged the automation industry with their simple to utilize platform wherever everybody with small otherwise no technical backdrop can start by discovering some necessary skills to program as well as run the Arduino board. These boards are used to extend separate interactive objects otherwise we can connect to software on your PC like MaxMSP, Processing, and Flash. This article discusses an introduction to Arduino mega 2560 board, pin diagram and its specifications.

The microcontroller board like “Arduino Mega” depends on the ATmega2560 microcontroller. It includes digital input/output pins-54, where 16 pins are analog inputs, 14 are used like PWM outputs hardware serial ports (UARTs) – 4, a crystal oscillator-16 MHz, an ICSP header, a power jack, a USB connection, as well as an RST button. This board mainly includes everything which is essential for supporting the microcontroller. So, the power supply of this board can be done by connecting it to a PC using a USB cable, or battery or an AC-DC adapter. This board can be protected from the unexpected electrical discharge by placing a base plate.

The SCL & SDA pins of Mega 2560 R3 board connects to beside the AREF pin. Additionally, there are two latest pins located near the RST pin. One pin is the IOREF that permit the shields to adjust the voltage offered from the Arduino board. Another pin is not associated & it is kept for upcoming purposes. These boards work with every existing shield although can adjust to latest shields which utilize these extra pins.

4.7.1 Arduino Mega Specifications

The specifications of Arduino Mega include the following.

- I. The ATmega2560 is a Microcontroller
- II. The operating voltage of this microcontroller is 5volts
- III. The recommended Input Voltage will range from 7volts to 12volts
- IV. The input voltage will range from 6volts to 20volts
- V. The digital input/output pins are 54 where 15 of these pins will supply PWM o/p.
- VI. Analog Input Pins are 16
- VII. DC Current for each input/output pin is 40 mA
- VIII. DC Current used for 3.3V Pin is 50 mA
- IX. Flash Memory like 256 KB where 8 KB of flash memory is used with the help of bootloader
- X. The static random access memory (SRAM) is 8 KB
- XI. The electrically erasable programmable read-only memory (EEPROM) is 4 KB
- XII. The clock (CLK) speed is 16 MHz
- XIII. The USB host chip used in this is MAX3421E
- XIV. The length of this board is 101.52 mm
- XV. The width of this board is 53.3 mm
- XVI. The weight of this board is 36 g

Arduino Mega Pin Configuration

The pin configuration of this **Arduino mega 2560** board is shown below. Every pin of this board comes by a particular function which is allied with it. All analog pins of this board can be used as digital I/O pins. By using this board, the Arduino mega projected can be designed. These boards offer flexible work memory space is the more & processing power that permits

Reset2 SCK MOSI MISO

GND MOSI MISO Vcc

USB to computer

7 to 12V DC input, center positive

ICSP for 16U2 USB interface

(I2C) SCL (I2C) SDA

SPI is not here

Interrupt 1 Interrupt 0

Interrupt 5 Interrupt 4 Interrupt 3 Interrupt 2

MISO SCK Reset

Vcc MOSI GND

5V pins 23 25 27 29

ICSP for ATmega2560

PWM on 44,45,46 (SPI) MOSI (SPI) SS use for digital ground

use for analog ground

not I2C not I2C UNO shields can cover up

reset

These pins are used for providing o/p regulated voltage approximately 5V. This RPS (regulated power supply) provides the power to the microcontroller as well as other components which are used over the Arduino mega board. It can be attained from Vin-pin of the board or one more regulated voltage supply-5V otherwise USB cable, whereas another voltage regulation can be offered by 3.3V0-pin. The max power can be drawn by this is 50mA.

The Arduino mega board includes 5-GND pins where one of these pins can be used whenever the project requires.

The RST pin of this board can be used for rearranging the board. The board can be rearranged by setting this pin to low.

The range of supplied input voltage to the board ranges from 7volts to 20volts. The voltage provided by the power jack can be accessed through this pin. However, the output voltage through this pin to the board will be automatically set up to 5V.

Serial Communication

The serial pins of this board like TXD and RXD are used to transmit & receive the serial data. Tx indicates the transmission of information whereas the RX indicates receive data. The serial pins of this board have four combinations. For serial 0, it includes Tx (1) and Rx (0), for serial 1, it includes Tx(18) & Rx(19), for serial 2 it includes Tx(16) & Rx(17), and finally for serial 3, it includes Tx(14) & Rx(15).

External Interrupts

The external interrupts can be formed by using 6-pins like interrupt 0(0), interrupt 1(3), interrupt 2(21), interrupt 3(20), interrupt 4(19), interrupt 5(18). These pins produce interrupts by a number of ways i.e. Providing LOW value, rising or falling edge or changing the value to the interrupt pins.

LED

This Arduino board includes a LED and that is allied to pin-13 which is named as digital pin 13. This LED can be operated based on the high and low values of the pin. This will give you to modify the programming skills in real time.

AREF

The term AREF stands for Analog Reference Voltage which is a reference voltage for analog inputs

Analog Pins

There are 16-analog pins included on the board which is marked as A0-A15. It is very important to know that all the analog pins on this board can be utilized like digital I/O pins. Every analog pin is accessible with the 10-bit resolution which can gauge from GND to 5

volts. But, the higher value can be altered using AREF pin as well as the function of analog Reference ().

I2C

The I2C communication can be supported by two pins namely 20 & 21 where 20-pin signifies Serial Data Line (SDA) which is used for holding the data & 21-pin signifies Serial Clock Line (SCL) mostly utilized for offering data synchronization among the devices

SPI Communication

The term SPI is a serial peripheral interface which is used to transmit the data among the controller & other components. Four pins like MISO (50), MOSI (51), SCK (52), and SS (53) are utilized for the communication of SPI.

Dimensions

The dimension of Arduino Mega 2560 board mainly includes the length as well as widths like 101.6mm or 4 inch X 53.34 mm or 2.1 inches. It is comparatively superior to other types of boards which are accessible in the marketplace. But, the power jack and USB port are somewhat expanded from the specified measurements.

Shield Compatibility

Arduino Mega is well-suited for most of the guards used in other Arduino boards. Before you propose to utilize a guard, confirm the operating voltage of the guard is well-suited with the voltage of the board. The operating voltage of most of the guards will be 3.3V otherwise 5V. But, guards with high operating voltage can injure the board.

In addition, the distribution header of the shield should vibrate with the distribution pin of the Arduino board. For that, one can connect the shield simply with the Arduino board & make it within a running state.

Programming

The programming of an Arduino Mega 2560 can be done with the help of an IDE (Arduino Software), and it supports C-programming language. Here the sketch is the code in the software which is burned within the software and then moved to the Arduino board using a USB cable.

An Arduino mega board includes a boot loader which eliminates an external burner utilization to burn the program code into the Arduino board. Here, the communication of the boot loader can be done using an STK500 protocol.

When we compile as well as burn the Arduino program, then we can detach the USB cable to remove the power supply from the Arduino board. Whenever you propose to use the Arduino board for your project, the power supply can be provided by a power jack otherwise Vin pin of the board.

Another feature of this is multitasking wherever Arduino mega board comes handy. But, Arduino IDE Software doesn't support multi-tasking however one can utilize additional operating systems namely RTX & FreeRTOS to write C-program for this reason. This is flexible to use in your personal custom build program with the help of an ISP connector.

Thus, this is all about an **Arduino Mega 2560 datasheet**. It is a substitution of the older Arduino Mega board. Because of the number of pins, usually, it is not utilized for general projects however we can discover them in complex projects such as temperature sensing, 3D printers, IOT applications, radon detectors, monitoring of real-time data applications, etc. Here is a question for you, what are the specifications of Arduino mega 2560 board?

4.8 NODEMCU ESP8266

NodeMCU is an open-source LUA based firmware developed for the ESP8266 wifi chip. By exploring functionality with the ESP8266 chip, NodeMCU firmware comes with the ESP8266 Development board/kit i.e. NodeMCU Development board. Since NodeMCU is an open-source platform, its hardware design is open for edit/modify/build.

NodeMCU Dev Kit/board consist of ESP8266 wifi enabled chip. The **ESP8266** is a low-cost Wi-Fi chip developed by Espressif Systems with TCP/IP protocol. For more information about ESP8266, you can refer to the ESP8266 WiFi Module.

There is Version2 (V2) available for NodeMCU Dev Kit i.e. **NodeMCU Development Board v1.0 (Version2)**, which usually comes in black colored PCB.

NodeMCU Dev Kit has **Arduino like** Analog (i.e. A0) and Digital (D0-D8) pins on its board.

It supports serial communication protocols i.e. UART, SPI, I2C, etc.

Using such serial protocols we can connect it with serial devices like I2C enabled LCD display, Magnetometer HMC5883, MPU-6050 Gyro meter + Accelerometer, RTC chips, GPS modules, touch screen displays, SD cards, etc. **NodeMCU with Arduino IDE**

Here is another way of developing NodeMCU with a well-known IDE i.e. Arduino IDE. We can also develop applications on NodeMCU using the Arduino development environment. This makes it easy for Arduino developers than learning a new language and IDE for NodeMCU.



Fig 4.11 NODEMCU ESP8266 DEVELOPMENT BOARD

4.8.1 ESP8266 Wi-Fi module specifications or features are given below.

- I. It is a powerful Wi-Fi module available in a compact size at a very low price.
- II. It is based on the L106 RISC 32-bit microprocessor core and runs at 80 MHz
- III. It requires only 3.3 Volts power supply
- IV. The current consumption is 100 m Amps
- V. The maximum Input/Output (I/O) voltage is 3.6 Volts.

- VI. It consumes 100 mA current
- VII. The maximum Input/Output source current is 12 mA
- VIII. The frequency of built-in low power 32-bit MCU is 80 MHz
- IX. The size of flash memory is 513 kb
- X. It is used as either an access point or station or both
- XI. It supports less than 10 microAmps deep sleep
- XII. It supports serial communication to be compatible with several developmental platforms such as Arduino
- XIII. It is programmed using either AT commands, Arduino IDE, or Lua script
- XIV. It is a 2.4 GHz Wi-Fi module and supports WPA/WPA2, WEP authentication, and open networks.
- XV. It uses two serial communication protocols like I2C (Inter-Integrated Circuit) and SPI (Serial Peripheral Interface).
- XVI. It provides 10- bit analog to digital conversion
- XVII. The type of modulation is PWM (Pulse Width Modulation)
- XVIII. UART is enabled on dedicated pins and for only transmission, it can be enabled on GPIO2.
- XIX. It is an IEEE 802.11 b/g/n Wi-Fi module with LNA, power amplifier, balun, integrated TR switch, and matching networks.
- XX. GPIO pins – 17
- XXI. Memory Size of instruction RAM – 32 KB
- XXII. The memory size of instruction cache RAM – 32 KB
- XXIII. Size of User-data RAM- 80 KB
- XXIV. Size of ETS systems-data RAM – 16 KB

4.9 ARDUINO IDE SOFTWARE

The Arduino IDE is an open-source software, which is used to write and upload code to the Arduino boards. The IDE application is suitable for different operating systems such as Windows, Mac OS X, and Linux. It supports the programming languages C and C++. Here, IDE stands for Integrated Development Environment.

The program or code written in the Arduino IDE is often called as sketching. We need to connect the Genuino and Arduino board with the IDE to upload the sketch written in the Arduino IDE software. The sketch is saved with the extension '.ino.'

Upload

The Upload button compiles and runs our code written on the screen. It further uploads the code to the connected board. Before uploading the sketch, we need to make sure that the correct board and ports are selected.

We also need a USB connection to connect the board and the computer. Once all the above measures are done, click on the Upload button present on the toolbar.

The latest Arduino boards can be reset automatically before beginning with Upload. In the older boards, we need to press the Reset button present on it. As soon as the uploading is done successfully, we can notice the blink of the Tx and Rx LED.

If the uploading is failed, it will display the message in the error window.

We do not require any additional hardware to upload our sketch using the Arduino Bootloader. A Bootloader is defined as a small program, which is loaded in the microcontroller present on the board. The LED will blink on PIN 13.

New

The New button opens the new window. It does not remove the sketch which is already present.

Open

It allows opening the sketch, which can be browsed from the folders and computer drivers.

Open Recent

The Open Recent button contains the list of the recent sketches.

Sketchbook

It stores the current sketches created in the Arduino IDE software. It opens the selected sketch or code in a new editor at an instance.

Examples

It shows the different examples of small projects for a better understanding of the IDE and the board. The IDE provides examples of self-practice.

Close

The Close button closes the window from which the button is clicked.

Save

The save button is used to save the current sketch. It also saves the changes made to the current sketch. If we have not specified the name of the file, it will open the 'Save As...' window.

Verify/Compile

It will check for the errors in the code while compiling. The memory in the console area is also reported by the IDE.

Upload

The Upload button is used to configure the code to the specified board through the port.

Upload Using Programmer

It is used to override the Bootloader that is present on the board. We can utilize the full capacity of the Flash memory using the 'Upload Using Programmer' option. To implement this, we need to restore the Bootloader using the **Tools-> Burn Bootloader** option to upload it to the USB serial port.

4.10 FIREBASE CLOUD

Google Firebase is a Google-backed application development software that enables developers to develop iOS, Android and Web apps. Firebase provides tools for tracking analytics, reporting and fixing app crashes, creating marketing and product experiment.

Firebase offers a number of services, including:

- I. Analytics – Google Analytics for Firebase offers free, unlimited reporting on as many as 500 separate events. Analytics presents data about user behavior in iOS and Android apps, enabling better decision-making about improving performance and app marketing.
- II. Authentication – Firebase Authentication makes it easy for developers to build secure authentication systems and enhances the sign-in and onboarding experience for users. This feature offers a complete identity solution, supporting email and password accounts, phone auth, as well as Google, Facebook, GitHub, Twitter login and more.
- III. Cloud messaging – Firebase Cloud Messaging (FCM) is a cross-platform messaging tool that lets companies reliably receive and deliver messages on iOS, Android and the web at no cost.
- IV. Realtime database – the Firebase Realtime Database is a cloud-hosted NoSQL database that enables data to be stored and synced between users in real time. The data is synced across all clients in real time and is still available when an app goes offline.
- V. Crashlytics – Firebase Crashlytics is a real-time crash reporter that helps developers track, prioritize and fix stability issues that reduce the quality of their apps. With crashlytics, developers spend less time organizing and troubleshooting crashes and more time building features for their apps.
- VI. Performance – Firebase Performance Monitoring service gives developers insight into the performance characteristics of their iOS and Android apps to help them determine where and when the performance of their apps can be improved.
- VII. Test lab – Firebase Test Lab is a cloud-based app-testing infrastructure. With one operation, developers can test their iOS or Android apps across a variety of devices.

4.10.1 Firebase use cases include:

- I. Create onboarding flows – developers can give users a quick, intuitive sign-in process using Firebase Authentication. They allow users to sign into their apps via their Google, Twitter, Facebook or GitHub accounts in less than five minutes. Developers can also track each step of their onboarding flows to enhance the user experience. Additionally, developers can use Google Analytics for Firebase to log events at each step of their onboarding flows, create funnels to determine where users are dropping off and use remote configuration to make changes to their apps to see how those changes affect conversions.
- II. Customize a “welcome back” screen – developers can use personalization to give every user the best experience by customizing the initial screen based on a user’s preferences, usage history, location or language. Developers can define audiences based, in part, on user behaviors and show targeted content to each audience.
- III. Progressively roll out new features – developers can launch new features with minimal risk by first testing those features on a few users to see how they work and how users respond. Then, when developers are satisfied, they can roll out their apps to the rest of their users.

Pricing

Firebase offers a free plan with 1 GB of real-time database storage and two paid subscription plans: Flame Plan (\$25 per month for 2.5 GB of storage) and Blaze Plan (pay-as-you-go, \$5 per GB of storage). All the plans include A/B testing, analytics, app indexing, authentication (except phone auth), cloud messaging, crashlytics, dynamic links, invites, performance monitoring, predictions and remote config.

The main differences between the plans include the allocated storage on the real-time database, number of upload operations, Cloud Firestore bandwidth, and more.

4.11 INSTALLATION OF REQUIRED LIBRARIES IN ARDUINO IDE SOFTWARE

ADD ZIP LIBRARIES

STEP – 1 : Click on sketch and then include library and select add. Zip library

STEP – 2 : Select the required libraries to be installed such as DHT , ArduinoJson, firebaseArduino.

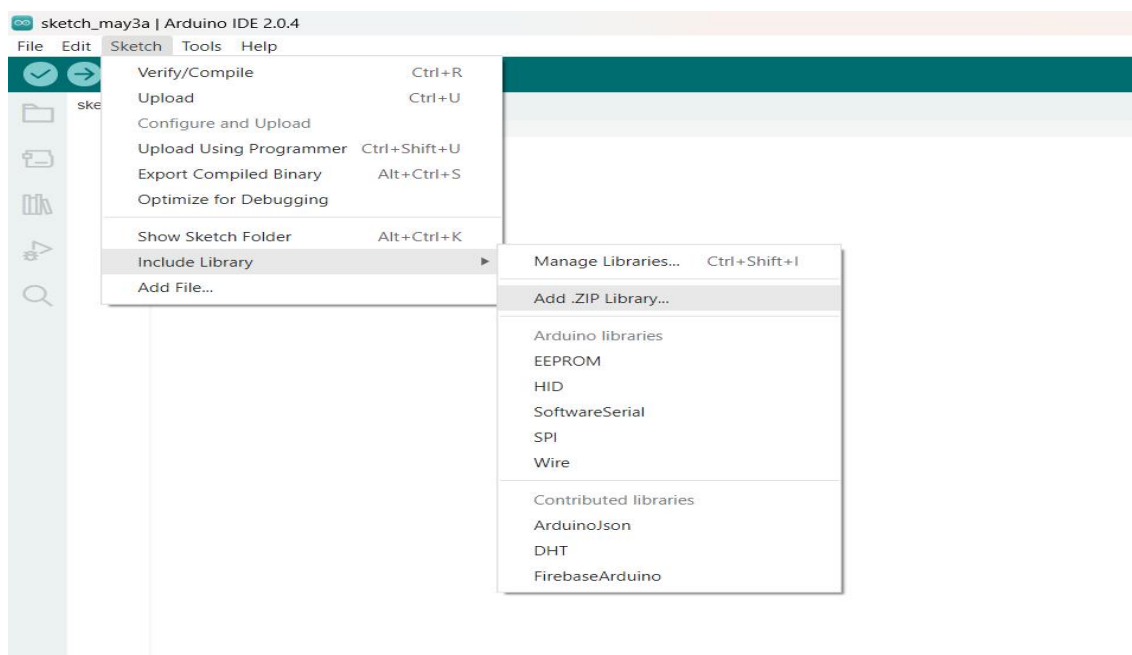


Fig 4.12 ADD ZIP LIBRARY

4.11.1 INSTALLATION OF ESP8266 BOARD

STEP – 1: Click on file and select preferences and paste the http link of esp package.

STEP – 2: Click on tools option and select Board Manager and search esp8266 and install the latest version of esp8266 boards.

STEP – 3: Click on tools option ,select Boards and again select esp8266 and click on NodeMCU 1.0 (ESP 12E-MODULE).

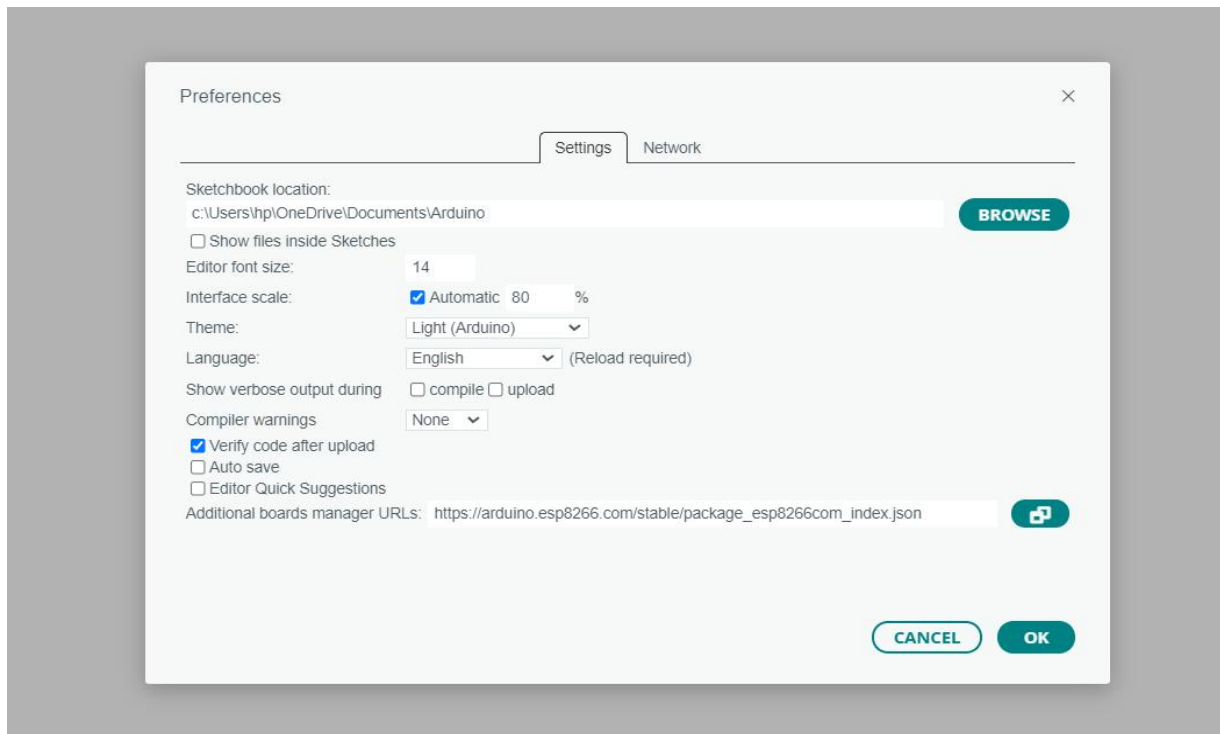


Fig 4.13 ESP BOARD PREFERENCES

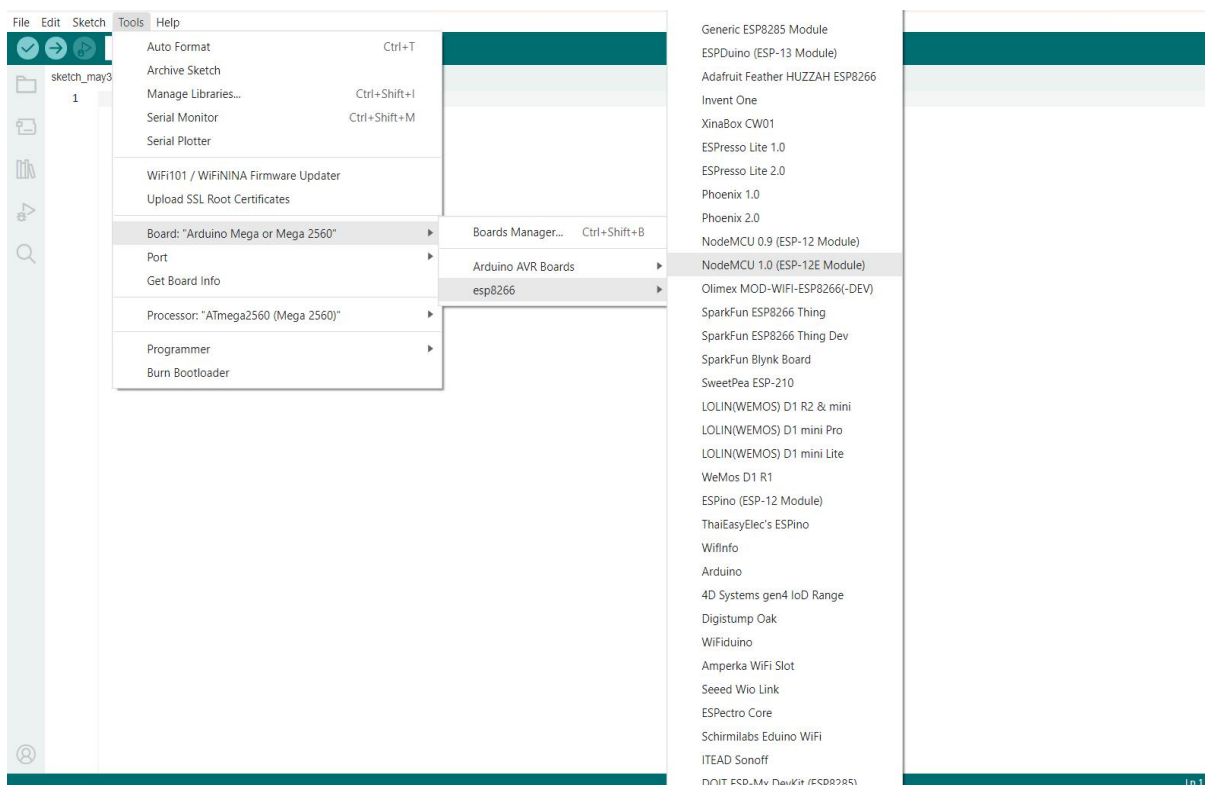


Fig 4.14. NODEMCU BOARD SELECTION

4.12 VOLTAGE MEASUREMENT USING ARDUINO

STEP – 1: The voltage sensor is connected parallel to the solar panel with positive and negative respectively.

STEP – 2: The Signal pin (S) of voltage sensor is connected to the A0 pin and (-ve) pin is connected to GND of Arduino Mega 2650.

STEP – 3: The voltage measurement code is written on the Arduino IDE software and uploaded.

4.12.1 SOURCE CODE

```
int value=0;
float voltage ;
float R1= 30000.0;
float R2= 7500.0;

void setup() {
  Serial.begin(9600);
}
void loop() {
  value = analogRead(A0);
  voltage= value*(5.0/1024)*((R1+R2)/R2);
  Serial.print("Voltage=");
  Serial.print(voltage);
  Serial.println("V");
  delay(1000);
}
```

4.13 CURRENT MEASUREMENT USING ARDUINO

STEP – 1: The current sensor is connected series to the solar panel with load.

STEP – 2: The Signal pin (S) of voltage sensor is connected to the A1 pin, VCC is connected to 5v and (-ve) pin is connected to GND of Arduino Mega 2650.

STEP – 3: The current measurement code is written on the Arduino IDE software and uploaded.

4.13.1 SOURCE CODE

```
void setup() {  
  Serial.begin(9600);  
}  
  
void loop() {  
  int adc = analogRead(A1);  
  float sensor = adc*5.0/1023.0;  
  float current = (sensor-2.50)/0.185;  
  
  if (current < 0.16){  
    current = 0;  
  }  
  Serial.print ("Current=");  
  Serial.print(current);  
  Serial.println("A");  
  delay(1000);  
}
```

4.14 LIGHT INTENSITY MEASUREMENT USING ARDUINO

STEP – 1: The LDR sensor is connected to the Arduino Mega 2560 directly.

STEP – 2: The Signal pin (S) of LDR sensor is connected to the A2 pin, VCC is connected to 5v and (-ve) pin is connected to GND of Arduino Mega 2650.

STEP – 3: The light intensity measurement code is written on the Arduino IDE software and uploaded.

4.14.1 SOURCE CODE

```
double calcLightIntensity(int AnVal, int Resistor){  
  
  double Vout= AnVal*0.0048828125;  
  int lux=((2500/Vout-500)/Resistor);  
  return lux;  
}  
void setup() {  
  Serial.begin(9600);
```

```

}

void loop() {
  int analogVal = analogRead(A2);
  int Resistance= 10;

  Serial.print("Light Intensity:");
  Serial.print(int(calcLightIntensity(analogVal, Resistance )));
  Serial.println(" lux");
  delay(1000);
}

```

4.15 TEMPERATURE AND HUMIDITY MEASUREMENT USING ARDUINO

STEP – 1: The DHT11 sensor is connected to the Arduino Mega 2560 directly.

STEP – 2: The Signal pin (S) of DHT11 sensor is connected to the A3 pin,VCC is connected to 5v and (-ve) pin is connected to GND of Arduino Mega 2650.

STEP – 3: The Temperature and Humidity measurement code is written on the Arduino IDE software and uploaded.

4.15.1 SOURCE CODE

```

#include "dht.h"
#define dht_apin A3
dht DHT;

void setup(){

  Serial.begin(9600);
  delay(500);
}

void loop(){
  DHT.read11(dht_apin);
  Serial.print("Current humidity = ");
  Serial.print(DHT.humidity);
  Serial.println("% ");
  Serial.print("temperature = ");
  Serial.print(DHT.temperature);
  Serial.println("C ");
  delay(1000);
}

```

```
}
```

4.16 SYNCHRONIZATION OF CODES

STEP – 1: All the codes are arranged in the sequential order.

STEP – 2: Verification of Analog pin number in similar to the source code.

STEP – 3: Upload the code.

4.16.1 SOURCE CODE

```
int offset =20;
int value=0;
float voltage ;
float R1= 30000.0;
float R2= 7500.0;
float power = 1.0;

double calcLightIntensity(int AnVal, int Resistor){

    double Vout= AnVal*0.0048828125;
    int lux=((2500/Vout-500)/Resistor);
    return lux;
}
#include "dht.h"
#define dht_apin A3

dht DHT;
void setup() {
    Serial.begin(9600);

}

void loop() {
    value = analogRead(A0);
    voltage= value*(5.0/1024)*((R1+R2)/R2);
    Serial.print("Voltage = ");
    Serial.print(voltage);
    Serial.println("V");
    delay(1000);

    int adc = analogRead(A1);
```

```

float sensor = adc*5.0/1024.0;
float current = (sensor-2.50)/0.185;
if (current < 0.16){
  current = 0;
}
Serial.print ("Current = ");
Serial.print(current);
Serial.println("A");
delay(1000);
// put your main code here, to run repeatedly:
power = voltage*current;
Serial.print("Power = ");
Serial.print(power);
Serial.println("W");
delay(1000);

int analogVal = analogRead(A2);
int Resistance= 10;

Serial.print("Light Intensity:");
Serial.print(int(calcLightIntensity(analogVal, Resistance )));
Serial.println(" lux");
delay(1000);

DHT.read11(dht_apin);

Serial.print("Current humidity = ");
Serial.print(DHT.humidity);
Serial.println("% ");
delay(1000);
Serial.print("temperature = ");
Serial.print(DHT.temperature);
Serial.println("C ");
delay(1000);
}

```

4.17 SERIAL COMMUNICATION

Serial communication is required when you want to transfer sensor data or any data from one device to another device, In our case it is ESP8266 NodeMCU and Arduino. Moreover, we will transfer Sensor data from Arduino to NodeMCU and NodeMCU to Arduino. Apart from this we will also see how to use software serial library on Arduino and Serial1 on ESP8266 NodeMCU, So that you can transfer data with another serial port TX RX, which we will be helpful.

4.17.1 SOURCE CODE

```
#include <SoftwareSerial.h>
void setup() {
  Serial.begin(9600);
}

void loop() {
  char buffer[60]="";

  if(Serial.available()>0)
  {
    Serial.readBytesUntil('\n',buffer,60);
    Serial.println(buffer);
  }
}
```

NOTE: UPLOAD THE ABOVE CODE TO NODEMCU ESP8266

Now the data is transferred from arduino to NodeMCU ESP8266 via Serial Communication and this data should be transmitted to the firebase cloud for monitoring purpose.

4.18 CREATING FIREBASE REALTIME DATABASE

STEP – 1: Go to firebase website and select (Go to console) .

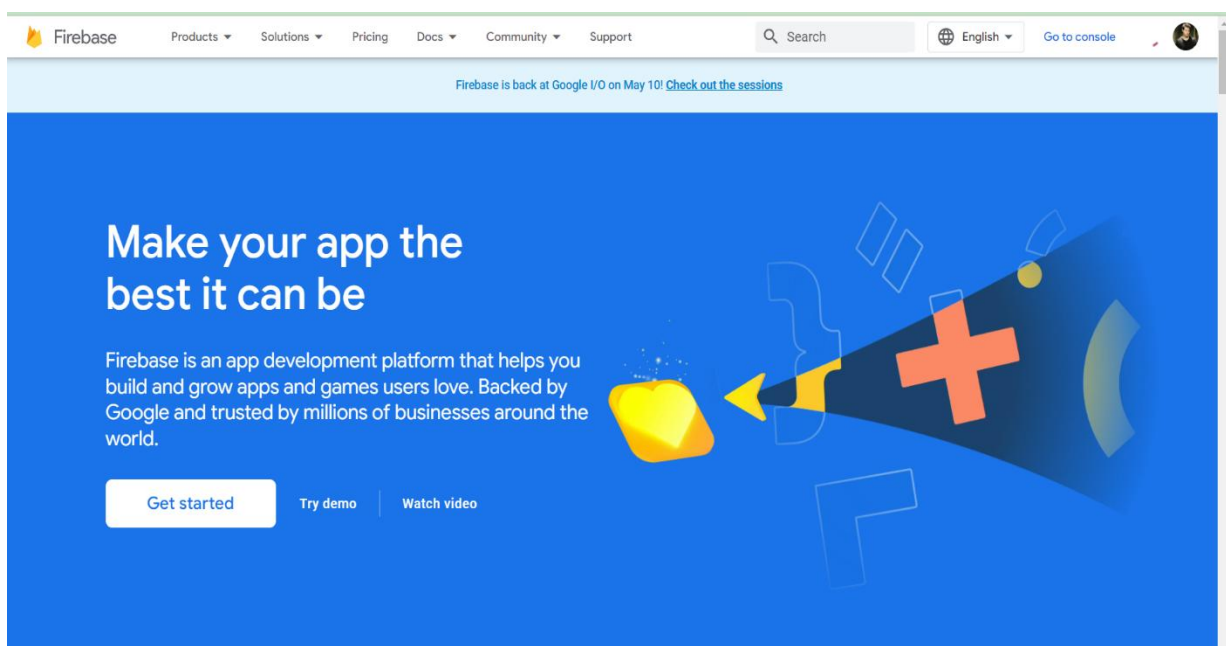


Fig 4.15 FIREBASE WEBSITE

STEP – 2: Select the project.

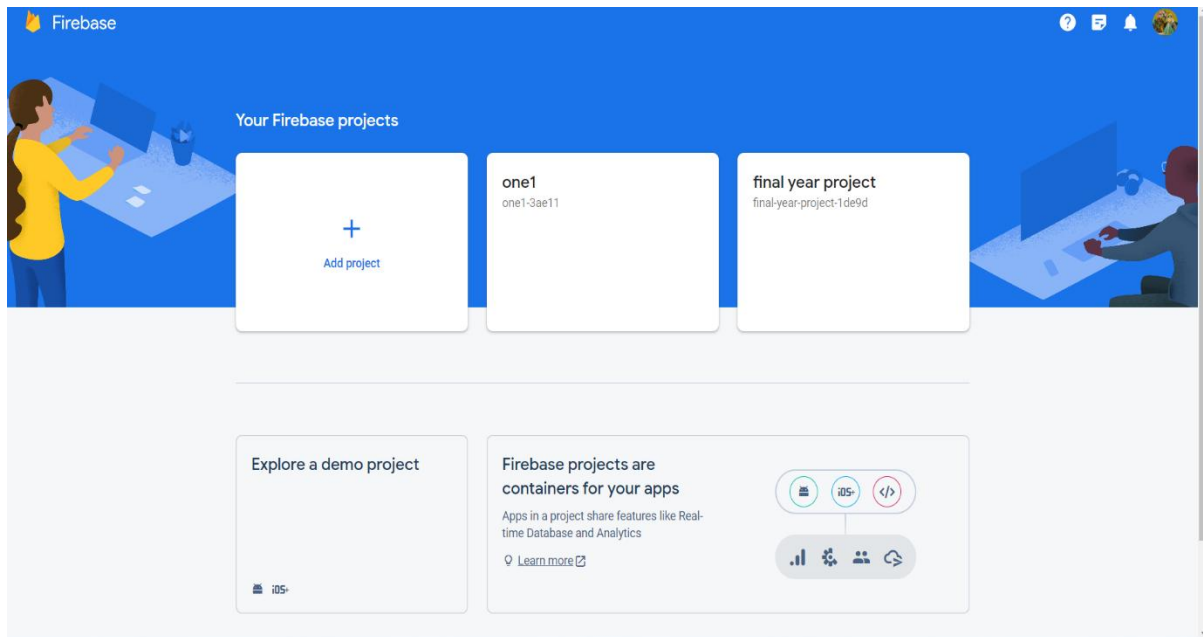


Fig 4.18.2 FIREBASE PROJECTS

STEP – 3: Go to project overview, select project settings, select service accounts and select database secrets.

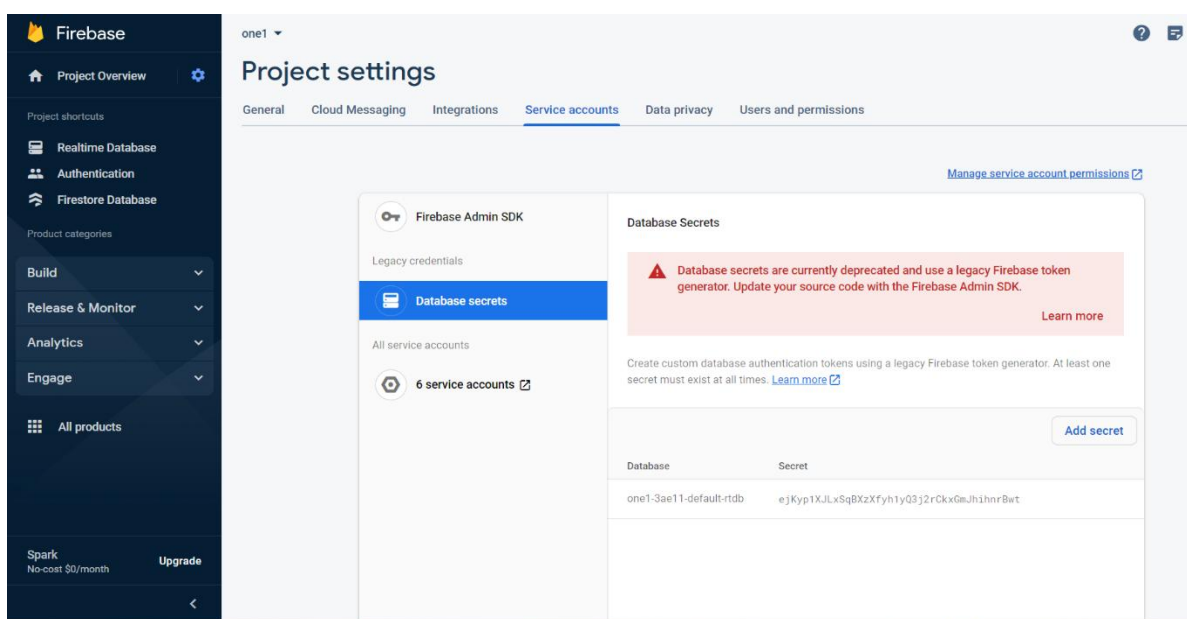


Fig 4.16 DATABASE SECRETES

STEP – 4: Next go to realtime database and select rules option and change the rules to true.

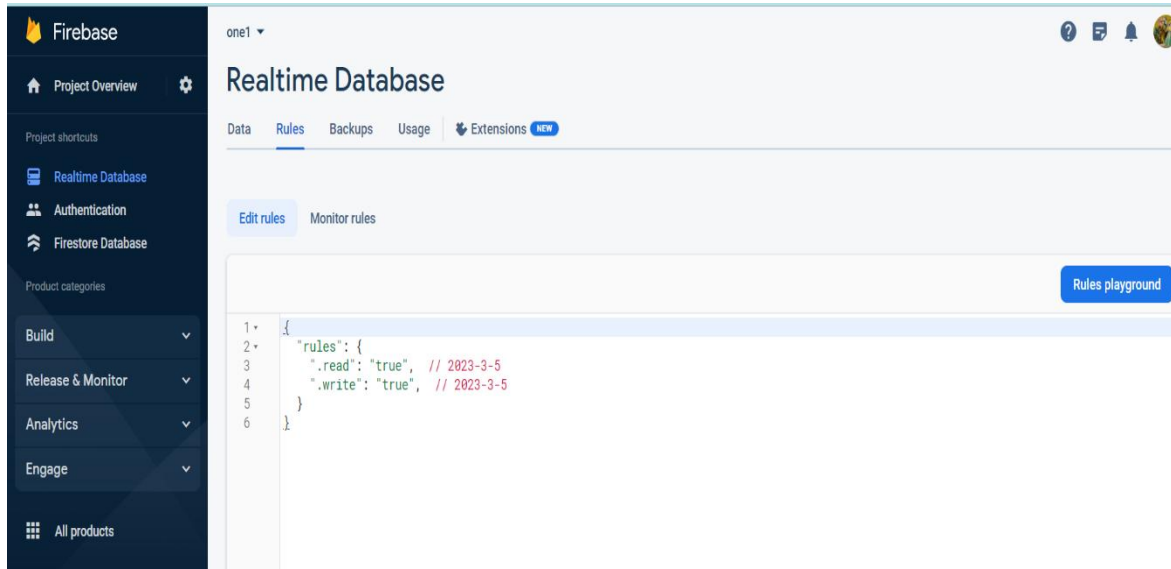


Fig 4.17 DATABASE RULES

STEP – 5: Realtime database is now able to receive the data from NodeMCU.

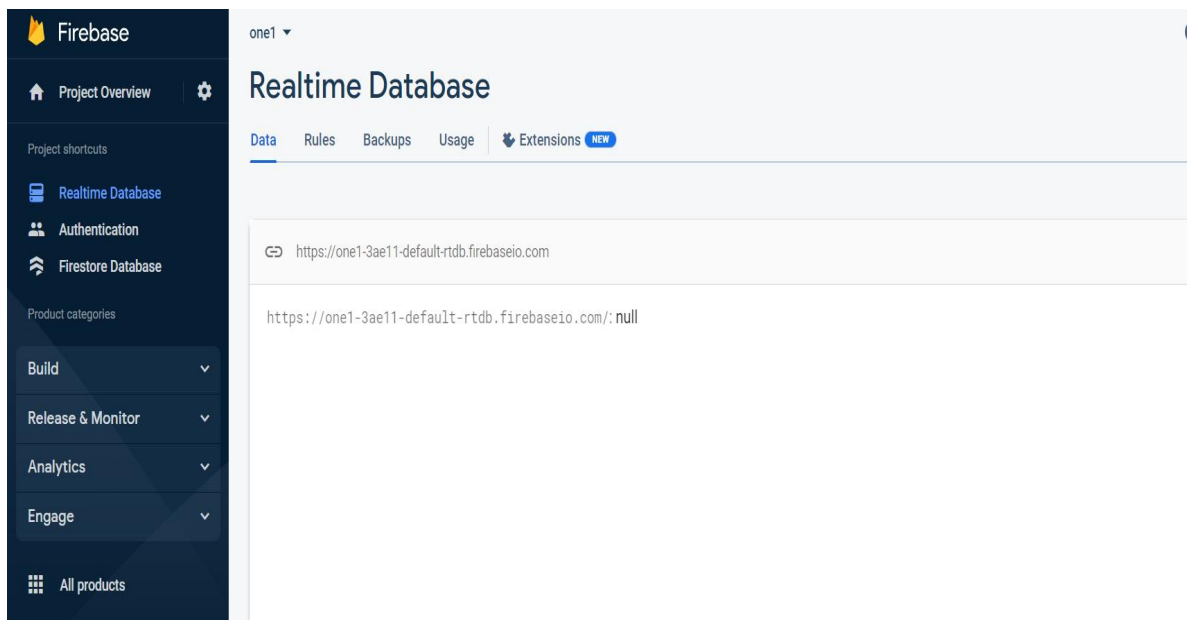


Fig 4.18 REALTIME DATABASE

4.18.1 UPLOADING DATA TO FIREBASE CLOUD

STEP – 1: Select the NodeMCU 1.0(ESP-12E MODULE) Board in Board Manager.

STEP – 2: Select the WIFI which is available with password.

STEP – 3: Include the FIREBASE HOST and FIREBASE AUTHENTICATION in the code.

STEP – 4: Include the Software Serial Library , FirebaseArduino.h Library, ESP8266WiFi.h Library in the code.

STEP – 5: Upload the code.

4.18.2 SOURCE CODE

```
#include <ESP8266WiFi.h>
#include <FirebaseArduino.h>
#include <SoftwareSerial.h>

// Set these to run example.
#define FIREBASE_HOST "one1-3ae11-default-rtdb.firebaseio.com"
#define FIREBASE_AUTH "ejKyp1XJLxSqBXzXfyh1yQ3j2rCkxGmJhlnrBwt"
#define WIFI_SSID "PETIT"
#define WIFI_PASSWORD "123456789"

void setup() {
  Serial.begin(9600);

  // connect to wifi.
  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
  Serial.print("connecting");
  while (WiFi.status() != WL_CONNECTED) {
    Serial.print(".");
    delay(500);
  }
  Serial.println();
  Serial.print("connected: ");
  Serial.println(WiFi.localIP());
```

```

Firebase.begin(FIREBASE_HOST, FIREBASE_AUTH);

}

void loop() {
  char buffer [60]="";
  if (Serial.available()>0){
    Serial.readBytesUntil('\n',buffer,60);
    Serial.println(buffer);
    String fire = String(buffer);

    // set string value
    Firebase.setString("PV PARAMETERS ",fire);
    // handle error
    if (Firebase.failed()) {
      Serial.print("setting /message failed:");
      Serial.println(Firebase.error());
      return;
    }
  }
}

```

4.19 IMPLEMENTATION OF REALTIME SYSTEM

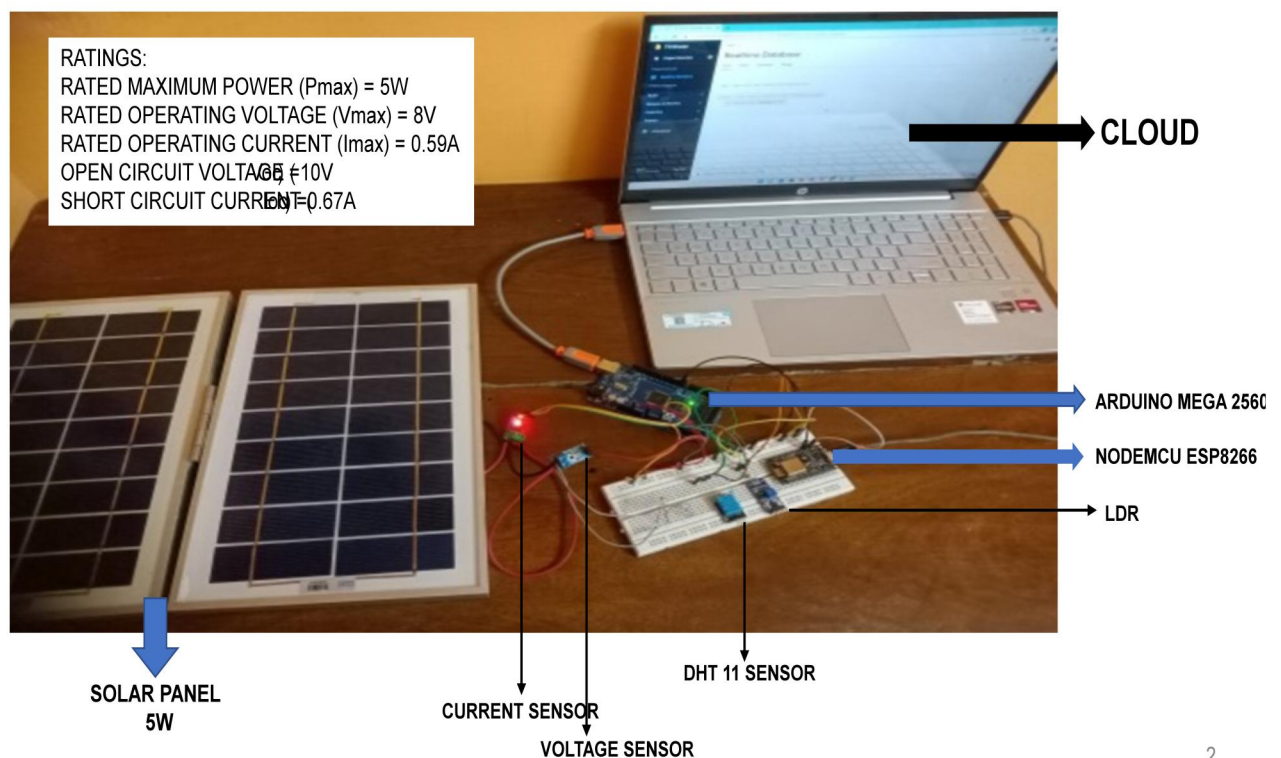


Fig 4.19 IMPLEMENTATION OF REALTIME SYSTEM

4.20 SUMMARY

The Implementation of IOT based PV monitoring system using ESP8266 consist of several stages which include testing the sensor on different conditions and various methods. After finalizing the effective method, the sensors are implemented for the specific task. The arduino starts working with the sensors and collects data. The arduino data are transferred to the NodeMCU ESP8266 via hardware connection and from the ESP8266 the data are send to the firebase cloud. For sending the data to the cloud we require WiFi with password and firebase authentication key and firebase project URL. From the firebase realtime database the datum are monitored effectively.

CHAPTER V

RESULTS AND DISCUSSIONS

5.1 REALTIME DATABASE RESULTS

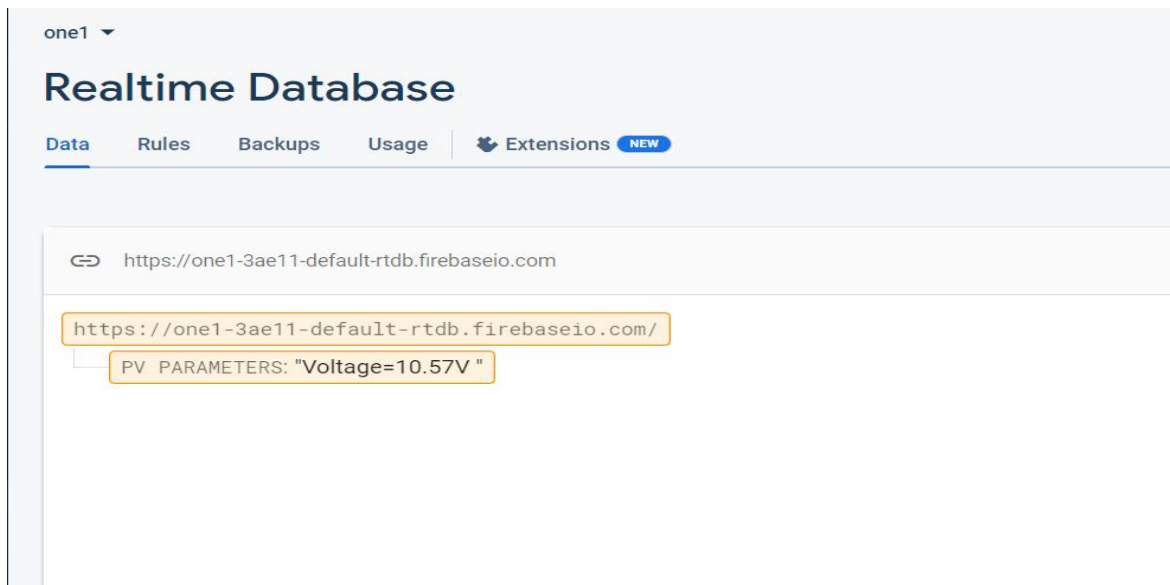


Fig 5.1 REALTIME VOLTAGE

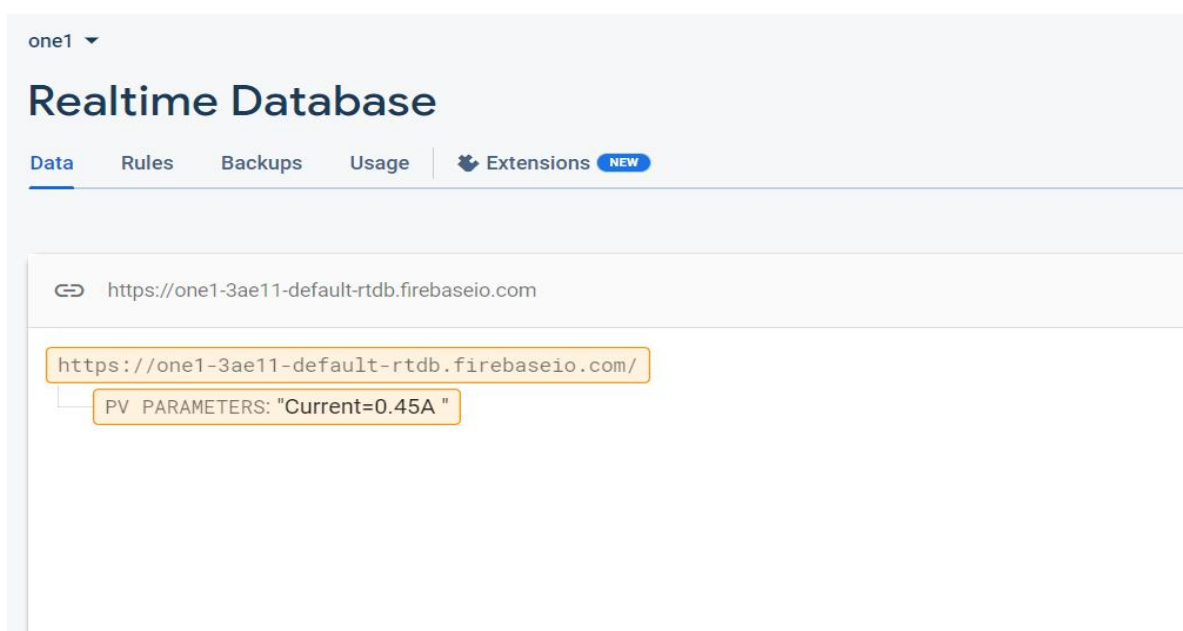


Fig 5.2 REALTIME CURRENT



Fig 5.3 REALTIME TEMPERATURE

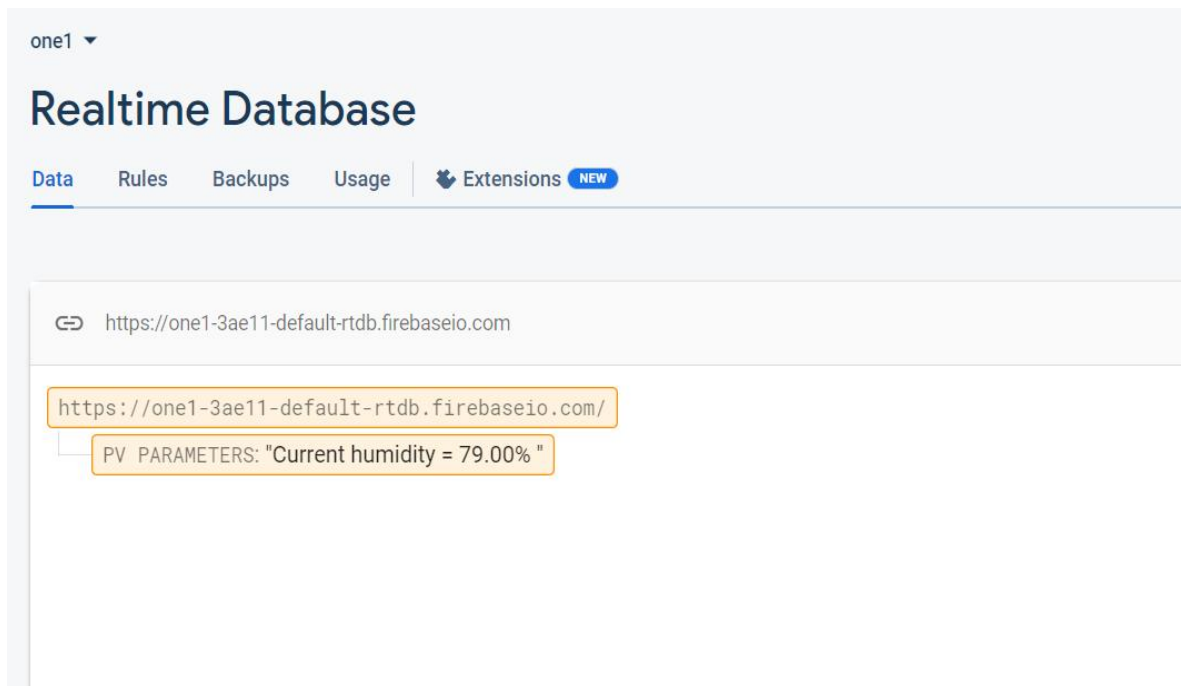


Fig 5.4 REALTIME HUMIDITY



Fig 5.5 REALTIME LIGHT INTENSITY

5.2 RESULTS ON A NORMAL DAY

5.2.1 FROM 9:00AM – 11:00AM

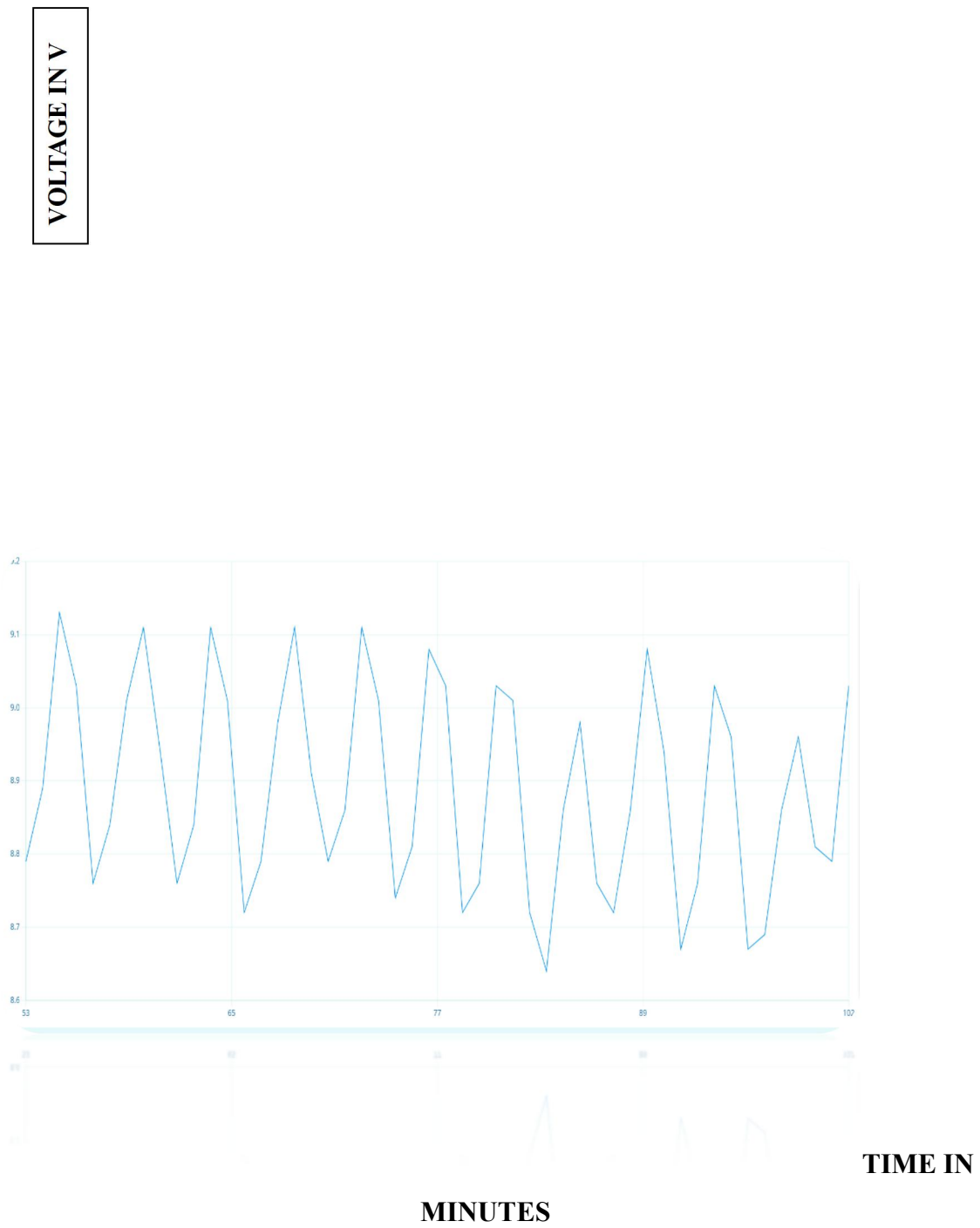
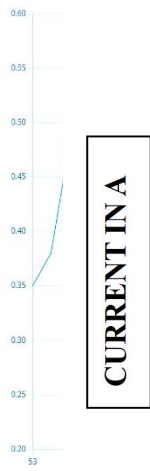


Fig 5.6 PERIODIC VOLTAGE



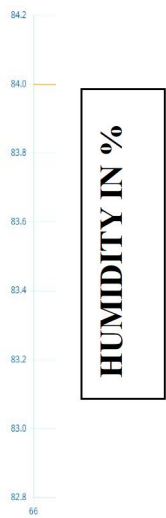
TIME IN MINUTES

Fig 5.7 PERIODIC CURRENT



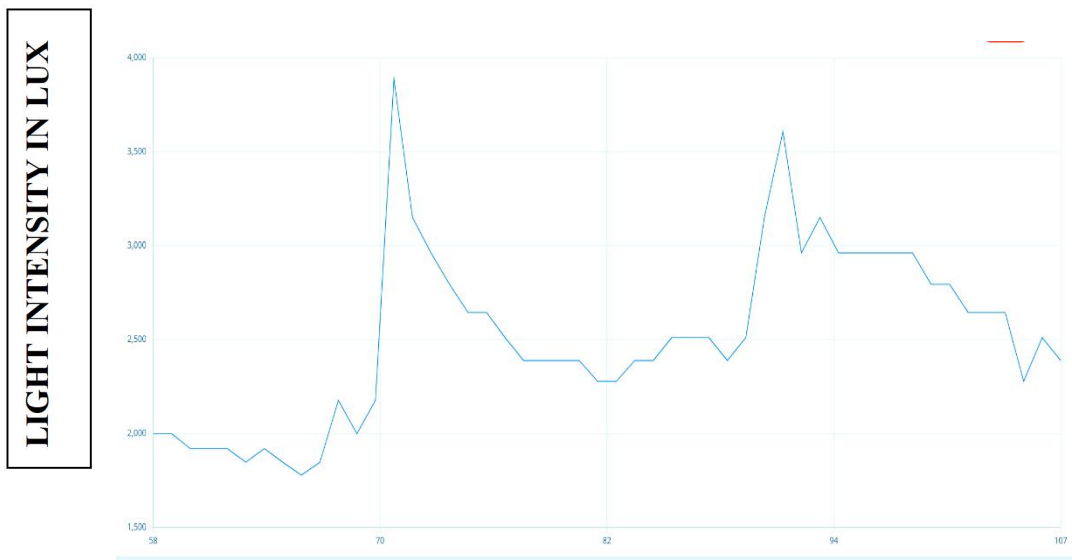
TIME IN MINUTES

**Fig 5.8 PERIODIC
TEMPERATURE**



TIME IN MINUTES

Fig 5.9 PERIODIC HUMIDITY



TIME IN MINUTES

Fig 5.10 PERIODIC LIGHT INTENSITY

5.2.2 FROM 11:00AM – 2:00PM

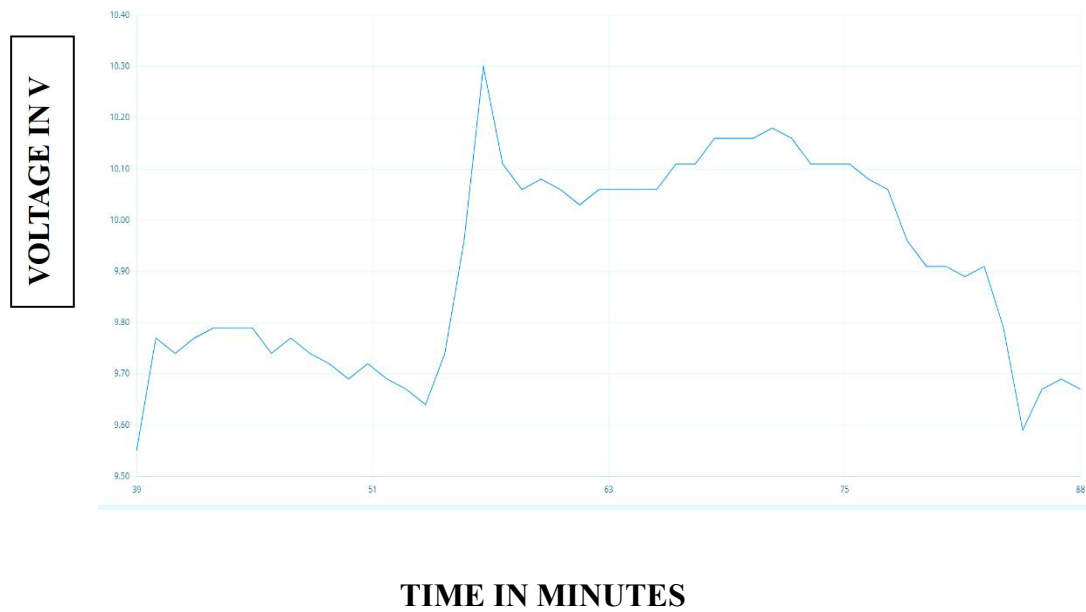


Fig 5.11 PERIODIC VOLTAGE

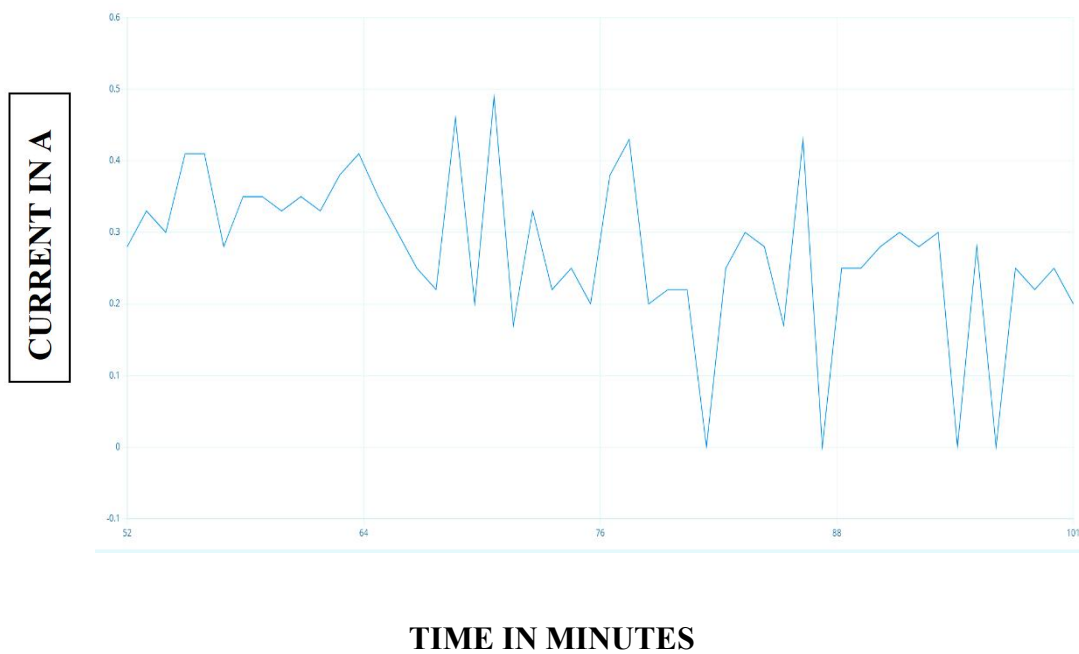


Fig 5.12 PERIODIC CURRENT

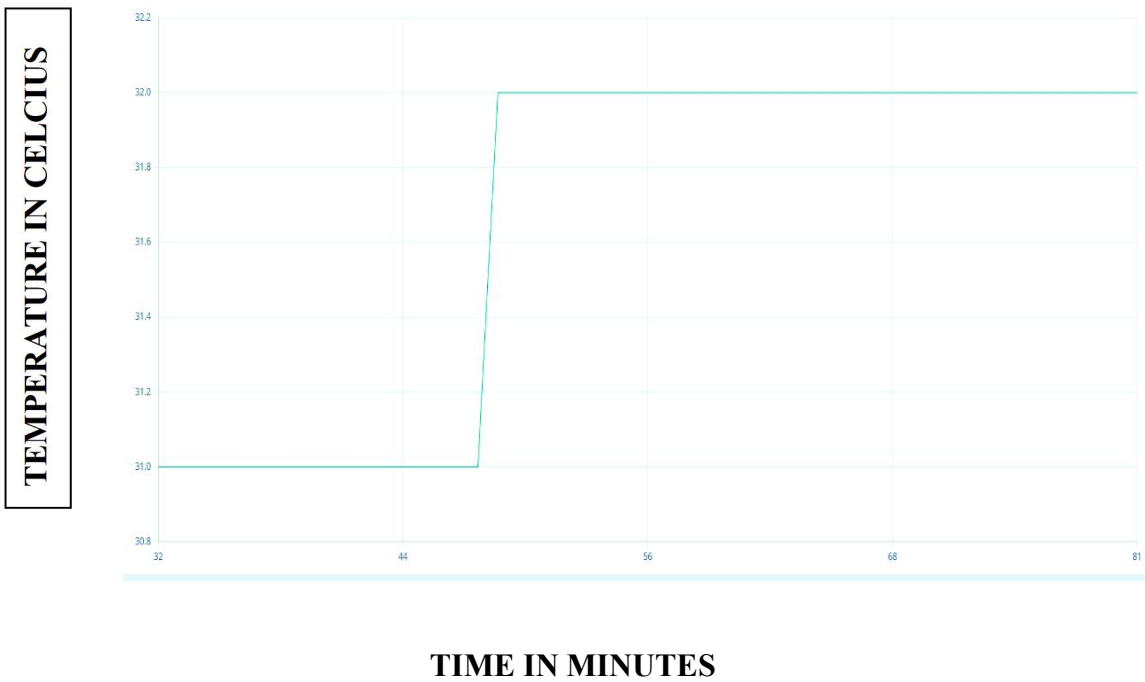


Fig 5.13 PERIODIC TEMPERATURE

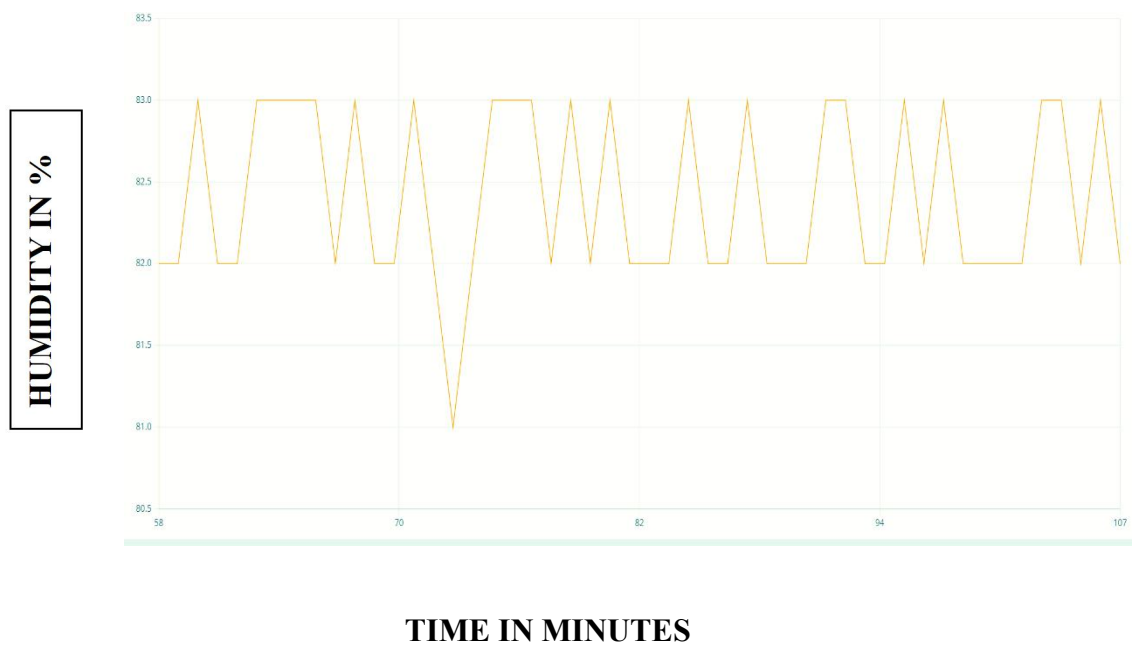


Fig 5.14 PERIODIC HUMIDITY

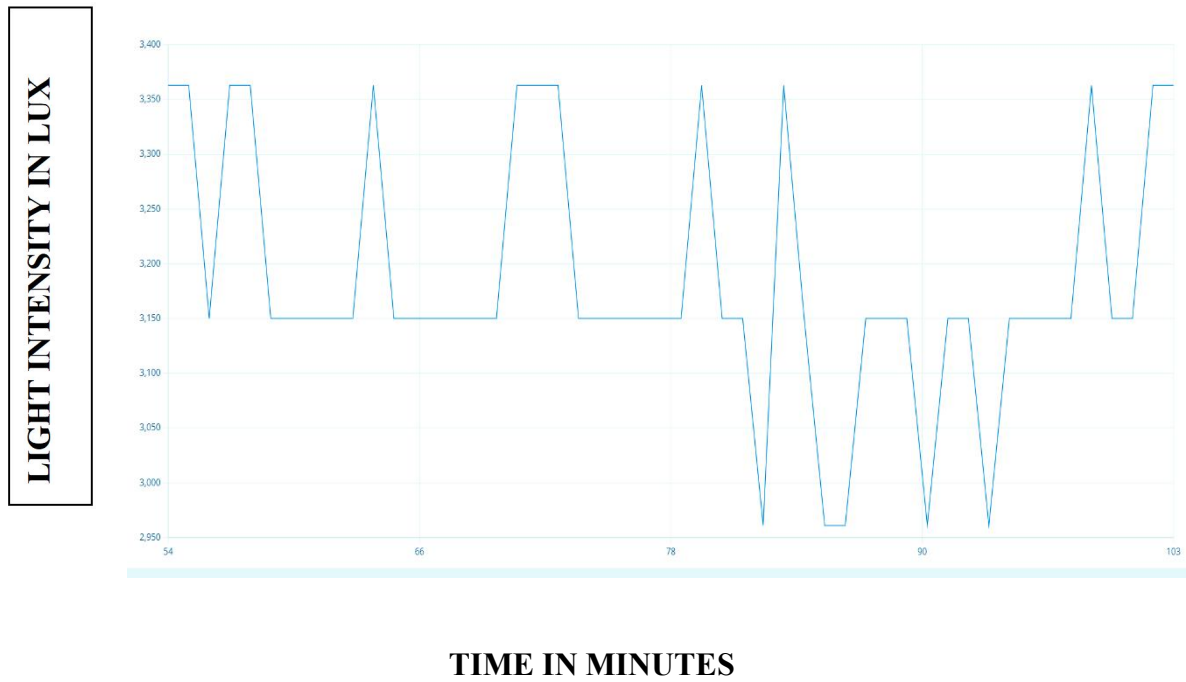


Fig 5.15 PERIODIC LIGHT INTENSITY

5.2.3 FROM 2:00PM – 4:00PM

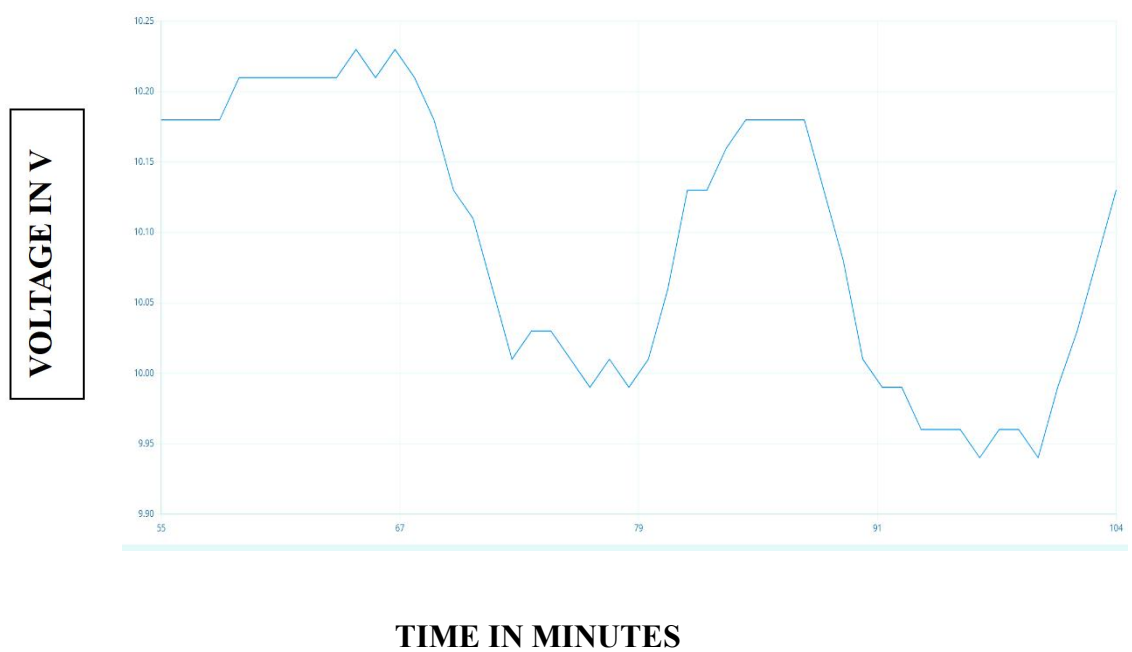


Fig 5.16 PERIODIC VOLTAGE

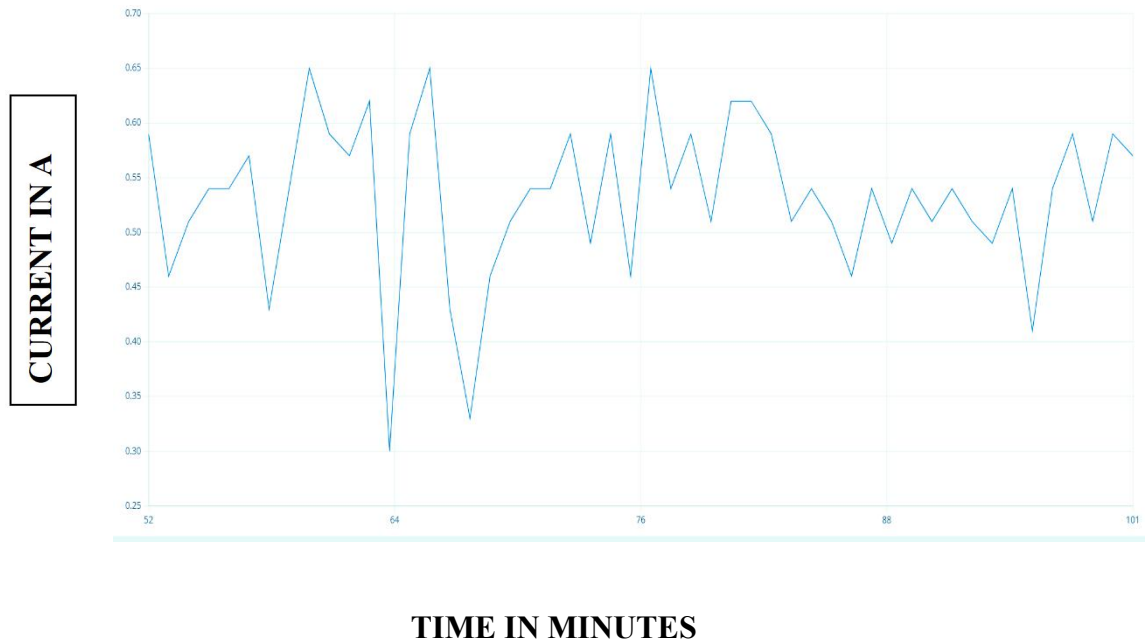


Fig 5.17 PERIODIC CURRENT

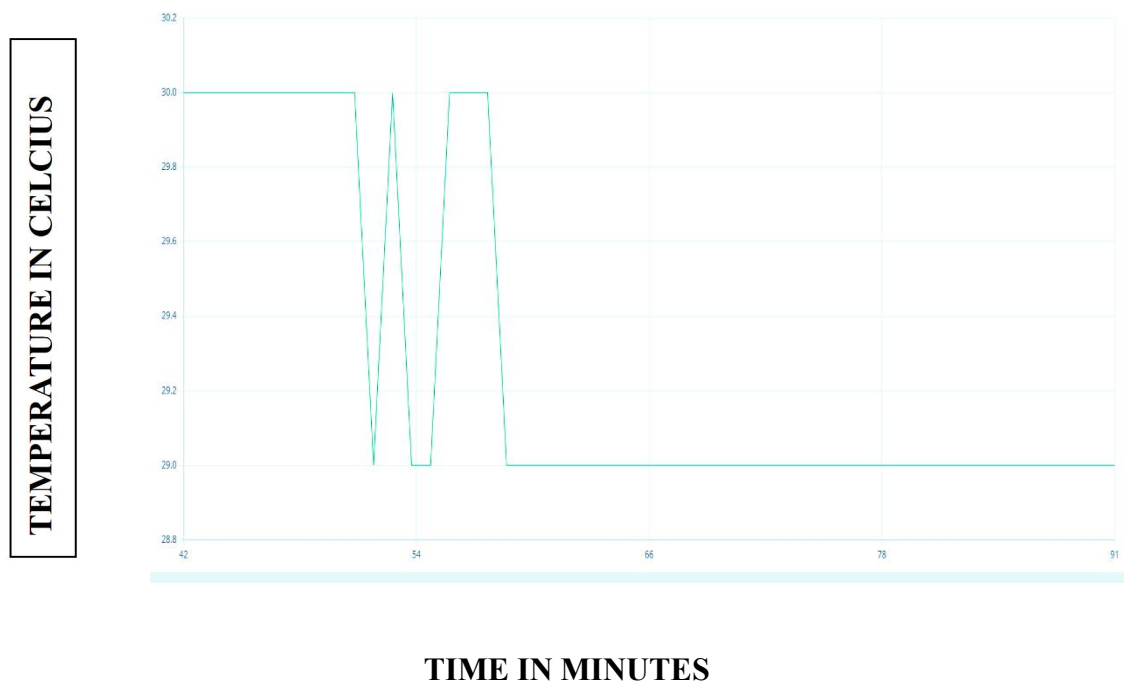


Fig 5.18 PERIODIC TEMPERATURE

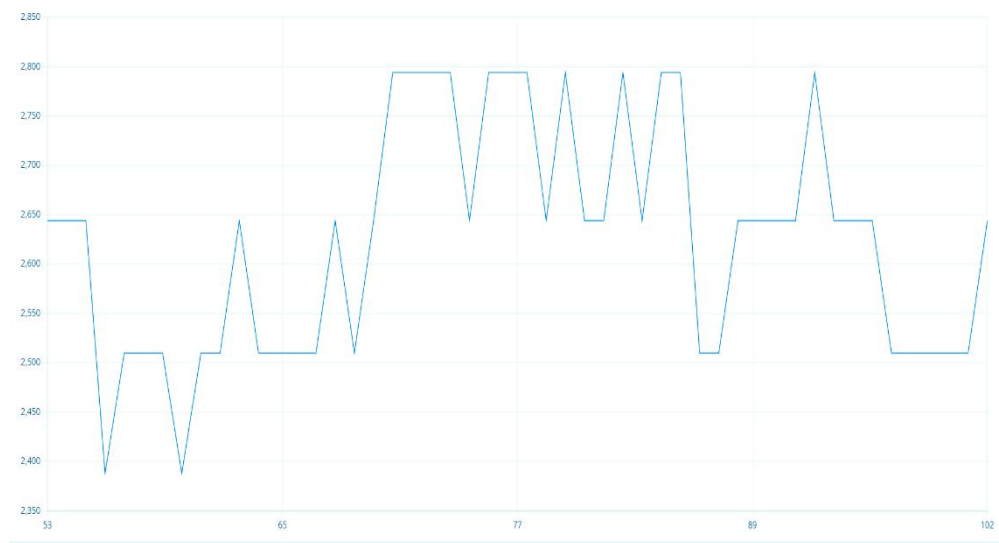
HUMIDITY IN %



TIME IN MINUTES

Fig 5.19 PERIODIC HUMIDITY

LIGHT INTENSITY IN LUX



TIME IN MINUTES

Fig 5.20 PERIODIC LIGHT INTENSITY

5.2.4 FROM 4:00PM – 6:00PM

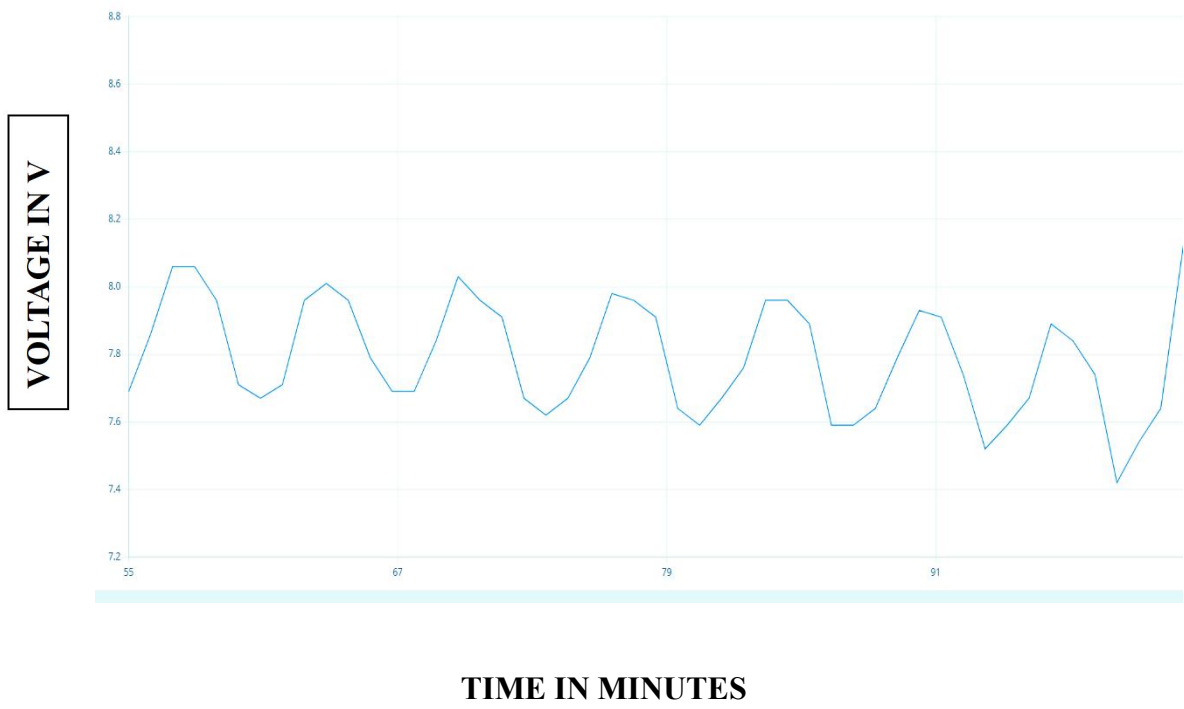


Fig 5.21 PERIODIC VOLTAGE

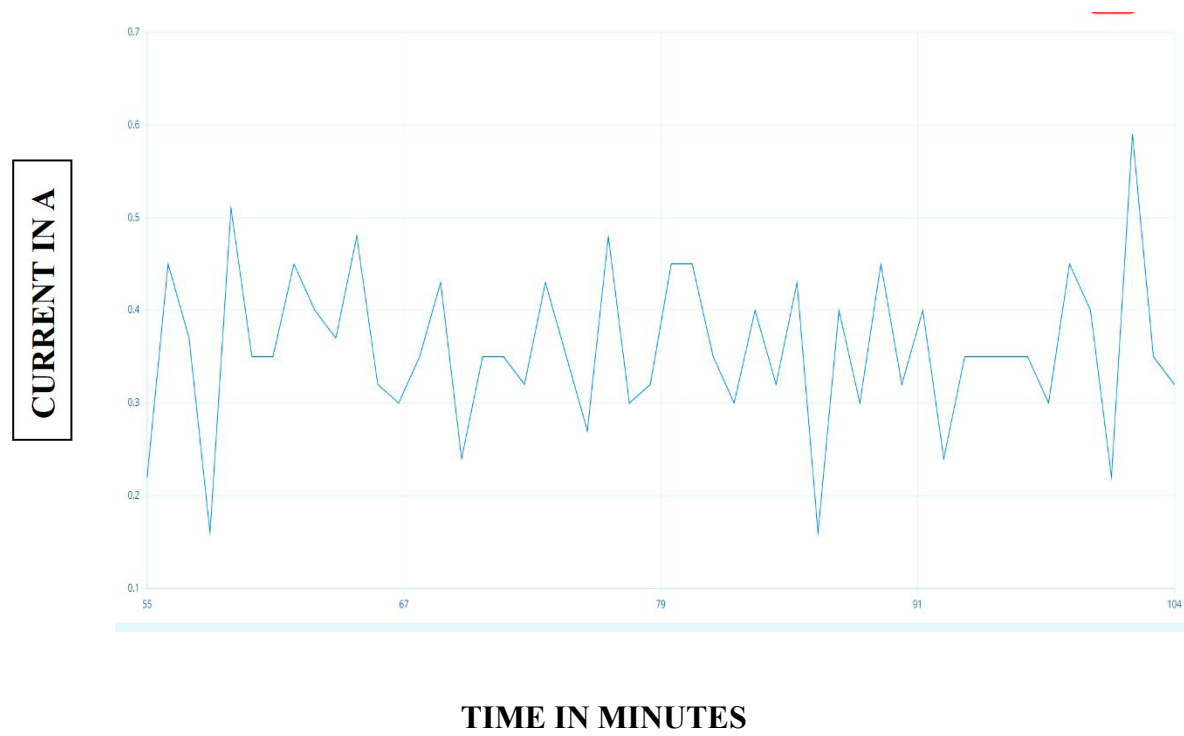


Fig 5.22 PERIODIC CURRENT

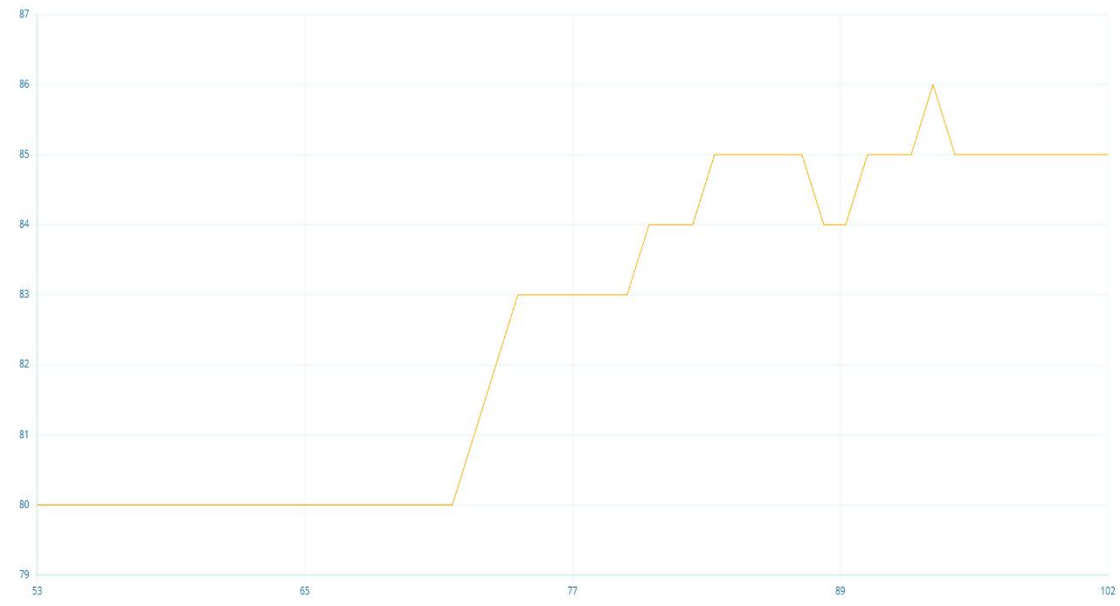
TEMPERATURE IN CELCIUS



TIME IN MINUTES

Fig 5.23 PERIODIC TEMPERATURE

HUMIDITY IN %



TIME IN MINUTES

Fig 5.24 PERIODIC HUMIDITY

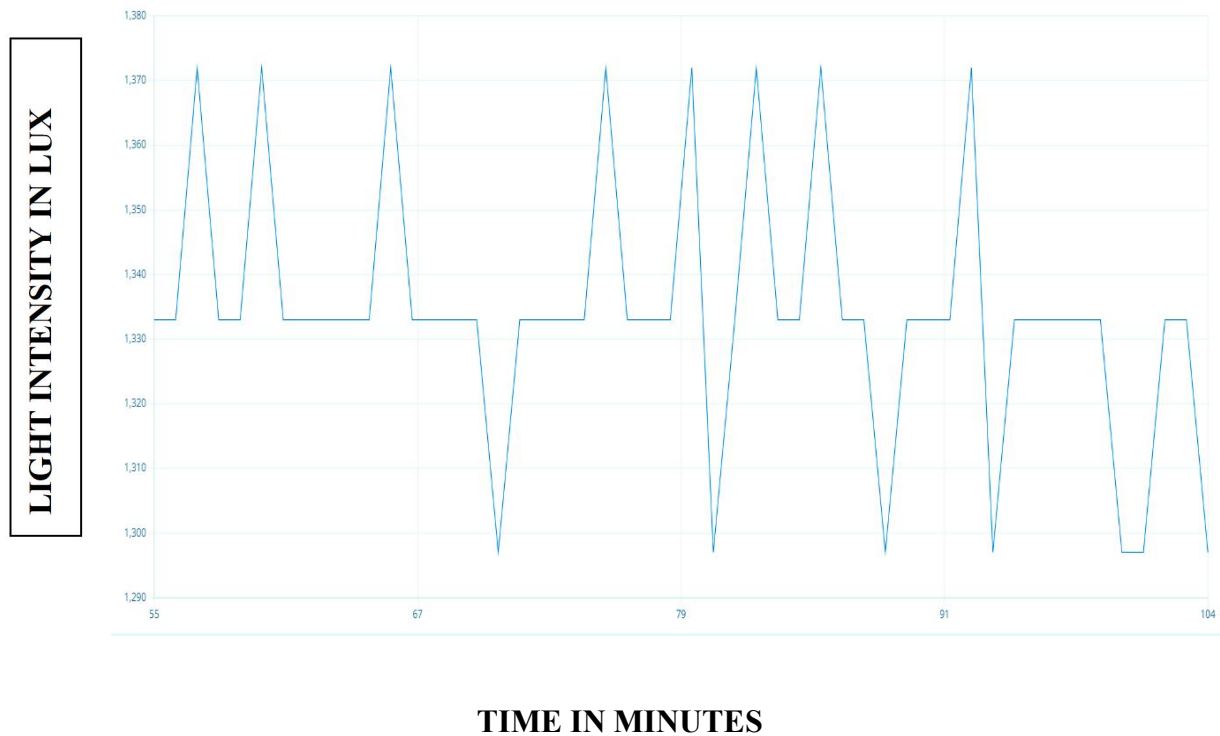


Fig 5.25 PERIODIC LIGHT INTENSITY

The above graphical results represent the PV parameters such as voltage, current, temperature, humidity and light intensity. These are the results which are taken periodically from morning to evening at separate interval of time. From morning 9:00AM – 11:00AM the results are Fig 5.6 to Fig 5.10 and from time 11:00AM – 2:00PM the results are Fig 5.11 to Fig 5.15 and from 2:00PM – 4:00PM the results are Fig 5.16 to Fig 5.20 and from time 4:00PM – 6:00PM the results are Fig 5.21 to Fig 5.25.

5.3 SUMMARY

The above chapter consists of several results which include firebase realtime database results and the arduino Serial Plotter results. The realtime database image represents the realtime results. These results are transferred to the arduino and processed to get accurate results and it is transferred to the nodeMCU and from nodeMCU it is sent to the cloud realtime database. The results which are in the Graphs are the same realtime results are plotted by the arduino IDE software by using the serial plotter .

CHAPTER VI

CONCLUSION

6.1 CONCLUSION

In the end, the solar monitoring gadget using IoT has been established efficaciously on this mission with the monitoring gadget for the power produced through the sun strength assets. With the implementation of IoT, it has eased the monitoring technique from every other location by using the Firebase utility, suitable for both Android and IOS users. This solar PV monitoring device sooner or later will assist the customers to evaluate and predict the overall performance of the solar panel. The users can determine what predictive maintenance needs to be finished to ensure the solar PV gadget function smoothly and can supply non-stop power to the load. There are numerous tips of works that can be proposed for a higher result in the future: so that you can enhance the specific percentage of

present-day analysis, it's miles recommended to degree the cutting-edge use of the clamp meter at some stage in the actual implementation of the solar PV machine in the fertigation machine. The proposed work helps to predict the performance of the Solar PV module through remote access. This can be extended for a large-scale solar plant to take preventive action by regularly monitoring the performance of the solar plant. It will be highly useful for the industrial and commercial application. The monitoring gadget ought to be examined in the actual setting of the farming so that that analyzing of the parameters wanted can be acquired effectively. It is better to construct or put into effect more accurate sensors so the fee may be displayed successfully. The accuracy may be set by way of calibrating the offset value of the sensors. For a better circuit method, it's far encouraged to design the circuit PCB to ease the relationship between the sensors with the microcontroller. Furthermore, with the PCB circuit, any blunders regarding the twine disconnection can be prevented.

As renewable energy sources become more integrated into the utility grid, using IOT to monitor a solar power plant is a crucial step. As a result, the automation and intellectualization of solar power plant monitoring will improve future decision-making for large-scale solar power plants and their grid integration. The approach is investigated, developed, and successfully achieved by the remote transmission of data to a Server for supervision through the web server. IoT-based remote monitoring will increase the system's energy efficiency by utilizing low-power sophisticated wireless modules, consequently lowering the carbon footprint. The use of a Web Console-based interface will greatly save manual supervision time and aid in the scheduling of plant management tasks.

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